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Effects of Continuous Neural Heterogeneity on Spiking Activity in a Spatially Embedded Network

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Abstract

This thesis investigates how continuously varying neuronal properties influence spiking activity in a spatially embedded spiking neural network. A one-dimensional recurrent network was implemented in NEST, where the membrane time constant and capacitance varied continuously across space.

Results show that temporally concentrated burst input preferentially activates neurons with shorter membrane time constants, whereas distributed input favours neurons with longer time constants. This temporal specialisation was dependent on input scaling: normalised input produced the clearest activity in the two temporal edge conditions, while intermediate burst conditions produced weak firing. Increasing the global input strength recruited more activity, but also changed the firing-rate profile of the distributed input condition.

These findings suggest that continuous intrinsic heterogeneity can generate input-sensitive temporal selectivity in spatially structured networks. The resulting activity patterns depend jointly on input structure, input magnitude, inhibition, and spatial organisation. The results should therefore be interpreted as a computational demonstration of input-sensitive temporal selectivity rather than as a robust biological prediction.

Keywords

Computational Neuroscience, Spiking Neural Networks, Spatial Embedding, Neural Heterogeneity, Leaky Integrate-and-Fire, Excitation and Inhibition

Sammanfattning

Denna avhandling undersöker hur kontinuerligt varierande neuronala egenskaper påverkar aktiviteten i ett spatialt inbäddat spikande neuralt nätverk. För detta ändamål implementerades ett endimensionellt nätverk i NEST, där membrantidskonstanten och kapacitansen varierade kontinuerligt längs den spatialla dimensionen.

Resultaten visar att tidsmässigt koncentrerad input främst aktiverar neuroner med kortare membrantidskonstanter medan utspridd input i högre grad gynnar neuroner med längre tidskonstanter. Denna temporala specialisering berodde dock på inputens styrka: normaliserad input gav de tydligaste aktivitetsmönstren i de två temporala randtillstånden, medan mellanliggande bursttillstånd gav svagare aktivitet. När den globala inputstyrkan ökades, aktiverades fler neuroner men samtidigt förändrades aktivitetsprofilen för det utspridda inputtillståndet.

Sammantaget tyder resultaten på att kontinuerlig neuronal heterogenitet kan bidra till inputkänslig temporal selektivitet i spatialt strukturerade nätverk. Samtidigt visar studien på att de resulterande aktivitetsmönstren beror på en samverkan mellan inputstruktur, inputstyrka, inhibition och spatial struktur. Resultaten bör därför främst tolkas som en beräkningsmässig demonstration av inputkänslig temporal selektivitet, snarare än som en direkt biologisk förklaring.

Nyckelbegrepp

Beräkningsneurovetenskap, spikande neurala nätverk, spatial inbäddning, neural heterogenitet, Leaky Integrate-and-Fire, excitering och inhibering

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Chapter 1

Introduction

Neural activity emerges from the interaction of many individual neurons, each receiving, integrating, and transmitting signals through synaptic connections. Understanding how such interactions give rise to larger-scale activity patterns is a central challenge in neuroscience. Since biological neural systems are highly complex, computational models provide a useful way to study specific mechanisms under controlled conditions.

Computational neuroscience uses mathematical and simulation-based models to investigate neural systems at different levels of abstraction [1], [2], [3], [4]. These models make it possible to explore how assumptions about neuronal properties, synaptic connectivity, and network structure influence neural activity. Depending on the research question, models may focus on detailed biophysical mechanisms, simplified neuron dynamics, or population-level network behaviour.

This thesis focuses on spiking neural networks, where neurons communicate through discrete spikes. Such models are well suited for studying how neuronal properties and network connectivity affect spiking activity over time. In particular, the work considers how spatial embedding and continuous neural heterogeneity may interact within a recurrent network. The following sections introduce the relevant background, define the specific problem addressed in the project, and describe the goal of the study.

1.1 Background

A growing area in computational neuroscience investigates how the incorporation of spatial structure and distance-dependent connectivity, called spatial embedding, shape neural dynamics [5], [6], [7]. Spatial embedding also raises questions about how excitation and inhibition interact across a network. At the same time, variation in neuronal properties, or neuronal heterogeneity, has received increasing attention, as it can influence spiking activity and firing-rate dynamics [8], [9]. Together, these areas motivate further investigation into how continuous gradients in neuronal properties affect activity in spatially structured networks.

1.2 Problem

Although spatial embedding and neural heterogeneity have both been studied separately, their combined effects remain less explored. Previous work has shown that distance-dependent connectivity can shape spiking activity [6], [7], and that intrinsic neuronal heterogeneity can influence firing-rate variability [8]. However, it remains unclear how continuously varying neuronal properties affect spiking activity within a spatially embedded network. This project therefore investigates the research question: What is the effect of continuously varying neural heterogeneity on spiking activity in a one-dimensional spatial network?

1.3 Goal

The goal of this degree project is to design and simulate a spatially embedded spiking neural network in which intrinsic neuronal properties vary continuously across the network. Specifically, the project implements a one-dimensional network of spiking neurons, introduces continuous heterogeneity in selected neuronal parameters, simulates spiking activity under different input conditions, and analyses the resulting activity patterns. The aim is to provide insight into how continuous gradients in neuronal properties influence activity in spatial neural systems.

1.4 Approach

The methodological approach of this study is quantitative and simulation-based. Rather than analysing experimental neural recordings directly, the project investigates neural dynamics through computational experiments in a controlled model environment. A spatially embedded spiking neural network is constructed, in which neurons are arranged along a one-dimensional spatial axis and assigned continuously varying intrinsic properties. By systematically varying the input structure and analysing the resulting spiking activity, the study examines how continuous neuronal heterogeneity influences network behaviour. This approach makes it possible to systematically vary selected model components, such as membrane properties, input timing, inhibition, and connectivity, and analyse how their interactions contribute to the resulting firing patterns. The simulations therefore provide a theoretical framework for exploring how spatial organisation and intrinsic neuronal variability may interact to shape spiking activity.

1.5 Delimitations

This project focuses on a simplified computational model of a spatially embedded spiking neural network. The neurons are modelled using simplified spiking neuron models, the *leaky integrate-and-fire* model, and do not incorporate all existing biophysical mechanisms of a neuron [2], [10], [11], [12]. Heterogeneity is introduced through a limited set of neuronal parameters, while other biologically relevant sources of variability are not considered.

The network structure is based on simplified spatial connectivity rules and does not aim to replicate specific biological brain architectures. In addition, the model is restricted to a one-dimensional representation, which is a simplified version of three-dimensional neural systems.

The size of the network and the duration of simulations are limited. The analysis focuses solely on spiking activity and does not address higher-level functions such as information processing or behaviour.

Finally, the results are not validated against experimental data and should therefore be interpreted within a purely computational and theoretical framework.

1.6 Outline

Chapter 1 introduces the background and motivation for the study, outlining the problem statement, research objectives, project scope, and overall structure of the work.

The theoretical foundations and terminology relevant to the project are presented in Chapter 2, including spiking neural networks, excitation and inhibition, spatial embedding, neural heterogeneity, and related research.

The methodology is described in detail in Chapter 3, including the experimental design, implementation, and construction of the conducted tests.

The experimental results and main findings of the study are presented in Chapter 4.

Chapter 5 discusses the findings, limitations, and broader implications of the study. Chapter 6 presents the main conclusions and outlines future work.

Chapter 2

Theoretical Background

This chapter presents the theoretical background and related work relevant to the project. Key concepts such as *spiking neural networks*, *spatially structured connectivity*, and *neural heterogeneity* are introduced, followed by a discussion of prior research and its relevance to this study.

2.1 Spiking Neural Networks

Spiking neural networks (SNNs) are models in which neurons communicate by electrical pulses, or spikes [10], [11]. This structure shares similarities with biological neurons [10], [13], where information is also transferred via spikes. In recent years, the interest in SNNs has increased [10], [11], [12]. One reason behind the increased interest is their potential to achieve performance comparable to currently used deep learning methods while being more energy efficient than them. [10], [11], [12], [14], [15].

Spike timing is assumed to be a significant factor in neural communication in SNNs [10], [11], as well as in biological neurons [10], [13]. Neurons in SNNs are commonly represented using simplified models such as the leaky integrate-and-fire (LIF) neuron [2], [10], [11]. SNNs are capable of capturing network activity and dynamics [2]. These properties make SNNs suitable for studying network dynamics.

However, despite their advantages, the accuracy of SNNs is still lower than traditional artificial neural networks [11]. Furthermore, many traditional SNN models simplify network structure, for example by ignoring spatial embedding.

2.1.1 Leaky Integrate-and-Fire Neuron

Different models of a biological neuron, such as the Hodgkin-Huxley model, integrate-and-fire model, as well as the leaky integrate-and-fire (LIF) differ in both computational efficiency and level of detail in their representations [2], [16]. The LIF neuron model is a simplified model of a biological neuron, and is commonly used for its computational efficiency [2], [10], [11], [12]. The LIF model collects (*integrates*) input spikes, and once the membrane potential of the neuron reaches the spiking threshold, the LIF neuron reaches the action potential and spikes (or *fires*). Once the neuron has spiked, the membrane potential is reset to the neuron's reset potential, where it is held for a refractory period before set to the neuron's resting potential, and integration resumes. In many cases, the reset and resting potential have the same value as the reset potential. The membrane potential V of each neuron evolves according to

$$\frac{dV}{dt} = -\frac{V - E_L}{\tau_m} + \frac{I_{\text{syn}}(t) + I_e}{C_m}\,, \tag{2.1}$$

where τ_m is the membrane time constant, E_L is the resting potential, I_{syn} is the synaptic input current, I_e is the constant input current, and C_m is the membrane capacitance [17].

The total synaptic input current is given by

$$I_{\text{syn}}(t) = I_{\text{syn,ex}}(t) + I_{\text{syn,in}}(t)\,, \tag{2.2}$$

where the excitatory and inhibitory synaptic currents are defined as

$$I_{\text{syn},X}(t) = \sum_j w_j \sum_k i_{\text{syn},X}(t - t_j^k - d_j)\,, \tag{2.3}$$

with for excitatory ($X = \text{ex}$) or inhibitory ($X = \text{in}$) synapses. j indexes the presynaptic neurons, k indexes the spike times of neuron j, and d_j denotes the synaptic delay from neuron j. The individual postsynaptic currents, $i_{\text{syn},X}(t)$ differ between different LIF implementations [18].

2.1.2 Neuron Properties

The behaviour of the LIF neuron model is determined by a set of parameters that describe the electrical properties of the neuron. These parameters determine how the neuron integrates incoming spikes and generates output spikes.

One important parameter is the membrane time constant τ_m, which determines how quickly the membrane potential decays toward the resting potential in the absence of input [15], [19]. A larger τ_m results in slower decay, allowing the neuron to integrate input over a longer time window while a smaller τ_m leads to faster decay.

The membrane capacitance C_m controls how strongly the membrane potential responds to input currents. τ_m is linearly dependent on C_m since $\tau_m = C_m \cdot R_m$, where R_m is the membrane resistance [2], [18], [19]. A larger capacitance results in slower changes in membrane potential, while a smaller capacitance allows for more rapid voltage changes.

The resting potential E_L represents the baseline voltage of the neuron. The membrane potential naturally decays toward this value over time. In addition, the neuron dynamics are influenced by the synaptic input current I_{syn} and the constant external input I_e, which together determine the total input to the neuron.

Finally, the firing threshold V_{th}, reset potential V_{reset}, and refractory period t_{ref} define the spiking behaviour of the neuron. When the membrane potential reaches V_{th}, a spike is emitted, after which the membrane potential is reset and held constant during the refractory period before normal integration resumes.

2.1.3 Excitation and Inhibition

When an excitatory neuron emits a spike, it induces a postsynaptic response in the neurons to which it is connected. The amplitude of this response is determined by the weight of the synaptic connection. In contrast, inhibitory neurons induce a decrease in the membrane potential of their postsynaptic targets when they emit spikes, thereby reducing their likelihood of firing.

Inhibitory connections play an important role in balancing network activity by counteracting excessive excitation and preventing uncontrolled increases in spiking activity. Such inhibitory mechanisms are also present in the neocortex [20] and are essential to stabilise neural dynamics **kilgore2024**, [21].

A ratio of approximately 20% inhibitory neurons and 80% excitatory neurons is commonly used in computational models and is consistent with cortical anatomy [7], [20], [22]. As shown in [7], the balance between excitation and inhibition has a strong influence on the overall activity of the network.

2.2 Spatially Embedded Networks

One important extension toward biological realism is the embedding of spatial structure in neural networks. Traditionally, many models of neural networks have disregarded explicit spatial embedding and instead assumed random connectivity [5], [6], [7], [23], [24]. However, these networks neglect biological constraints associated with long-range wiring. Spatially structured networks incorporate the physical positions of neurons. In such networks, connection probabilities between two neurons typically decrease with distance, reflecting the biological costs associated with long-range wiring [5], [6]. Importantly, spatial embedding is not only a structural detail but plays a central role in neural networks, as it influences neural dynamics and behaviour [5], [23], [24].

2.3 Neural Heterogeneity

In addition to spatial embedding, another key aspect of biological realism is neural heterogeneity [12], [25]. Neural heterogeneity arises from varying properties within a neural network. The variability can occur in both the cellular properties of neurons, for example the membrane time constant and the spiking threshold, as well as in the network through differences in synaptic weights and connectivity structure [8]. Networks incorporating intrinsic neuronal heterogeneity are considered more biologically realistic [8], [25], [26], [27] and have been shown to influence spiking activity, including firing rates [8], [25], [28], [29].

2.4 Related Work

2.4.1 Spatially Embedded Spiking Neural Networks

Spatial embedding has been shown to influence the dynamics of spiking neural networks (SNNs). [6] demonstrated that spatial embedding affects spiking activity through heterogeneous connectivity patterns, such as excitatory and inhibitory connections. Their results showed that spatially embedded networks exhibit spatio-temporal propagation of activity that does not emerge in randomly connected networks. However, intrinsic neuronal heterogeneity was not considered in their study.

Similarly, [23] investigated how different connectivity schemes, ranging from random to spatially dependent, influence spiking behaviour. In addition, they examined how neuron placement, from random to regular spatial distributions, affects network dynamics. Their findings suggest that spatial embedding plays a significant role in shaping network behaviour through both connectivity structure and neuronal positioning.

[5] discussed the principles underlying spatially embedded networks and argued that such architectures provide a more biologically realistic representation of neural systems. They further concluded that spatial

embedding substantially influences network dynamics. More recently, [24] examined how spatial embedding affects the performance of SNNs by comparing networks with varying degrees of spatial organization, ranging from non-embedded to fully three-dimensional implementations. Their results showed that spatially embedded networks outperformed networks without spatial embedding.

Together, these studies highlight the importance of spatial constraints in neural networks. However, these studies primarily focus on spatial embedding and do not investigate continuous intrinsic neuronal heterogeneity within spatially embedded networks.

2.4.2 Neural Heterogeneity in Spiking Neural Networks

Neural heterogeneity has also been shown to impact the behaviour and performance of SNNs. [25] investigated heterogeneity in neuron parameters and found that it can improve network performance and influence spiking dynamics, suggesting that variability in neuron properties enhances computational capabilities.

Furthermore, [12] examined the role of the membrane time constant and showed that it shapes distinct spiking dynamic regimes under temporally concentrated versus temporally distributed input. This work introduces intrinsic neuronal heterogeneity by assigning neuron-specific membrane time constants and demonstrates its importance in shaping network activity. However, it does not consider continuous intrinsic neuronal heterogeneity, nor does it employ a spatially embedded network.

[9] investigated the role of heterogeneity by varying spike thresholds in spiking neural networks. They found that heterogeneity influences both neural dynamics and computational function. However, their study does not consider spatially structured connectivity, instead focusing on the functional role of heterogeneity in non-spatial network models.

Overall, while previous work has studied spatial embedding and neural heterogeneity separately, the combined effect of continuously varying neuron properties in spatially structured spiking neural networks

remains less explored. This gap is the focus of the present project.

Chapter 3

Methods

3.1 Simulation Framework

All simulations were implemented using `NEST` version 3.9.0 [17], a Python-based simulator for studying networks of neurons. `NEST` was chosen for its scalability and built-in support for spatial networks. Numerical computations and data processing were performed using `NumPy`. Figures were generated using `Matplotlib`.

The simulations were organised into four stages. First, single-neuron and uncoupled-population simulations were used to validate the selected neuron parameters and the continuous membrane-time-constant gradient. Second, input-generation experiments were performed to evaluate the effect of temporal jitter, stochastic transmission and background noise. Third, simplified network configurations were used to compare externally driven activity with internally generated recurrent activity. Finally, the full connected heterogeneous network was simulated to evaluate how continuously varying neuronal properties shaped spiking activity across the spatial gradient.

Experiment	Network size	Duration (ms)	Main variables	Output measures
Two-neuron validation	2	1000	τ_m	Peak membrane potential
Population validation	76	1000	τ_m	Membrane potential trace
Input normalisation	100	1000	Number input spikes	Peak membrane potential
Input jitter & stochastic transmission	100	2000	Bursts, σ_{gen}, p_{tx}	Firing activity
Background noise	100	2000	Noise	Firing activity
External/recurrent comparison	100	2000	No connections / E-only / E & I	Spike count, drive ratio
Final heterogeneous network	1000	2000	Bursts & input strength regimes	Firing rate and spike timing

Table 3.1.1: Summary of simulation experiments, network sizes, durations, variables, and output measures. All simulations used a resolution of 0.1ms.

3.2 Models and parameters

The spiking dynamics of the network is strongly dependent on the choice of neuron model and the parametrisation of neuronal and synaptic properties.

3.2.1 Neuron Model

The LIF model was implemented using the `iaf_psc_exp` neuron model in NEST, in which synaptic input currents are modelled as exponentially decaying functions [17]. The membrane potential V of each `iaf_psc_exp` neuron evolves according to Eq. (2.1) presented in Section 2.1.1. The synaptic input current I_{syn} is composed of postsynaptic currents shaped exponentially as defined in Eqs. (2.2) and (2.3). For a `iaf_psc_exp` neuron, the individual postsynaptic current is given by

$$i_{\text{syn},X}(t) = w \cdot e^{-\frac{t}{\tau_{\text{syn,X}}}} \cdot \Theta(t) \tag{3.1}$$

where w is the connection weight that determines the amplitude of the postsynaptic current, $\tau_{\text{syn},X}$ is the synaptic time constant, with $X \in \{\text{ex}, \text{in}\}$ [18]. $\Theta(t)$ denotes the Heaviside step function. The Heaviside function ensures that the postsynaptic current is zero for $t < 0$, such

that the synaptic response begins only after the arrival of a presynaptic spike [18].

Alternative neuron models available in `NEST` were evaluated, including `iaf_psc_delta`, which models synaptic input as instantaneous current pulses. However, the `iaf_psc_exp` model was selected for all simulations because its exponentially decaying postsynaptic currents provide a more biologically plausible representation of synaptic dynamics while maintaining low computational cost.

3.2.2 Neuron Parameters

The membrane time constant was selected as the primary gradient heterogeneous parameter due to its direct influence on the temporal integration of synaptic inputs; see Eq. (2.1), making it well-suited to investigate how continuous gradients in neuronal properties affect spiking activity. For all experiments, the membrane time constant was linearly varied from 5 to 80 ms.

In this model, the membrane capacitance for all neurons was parametrised as $C_m = \frac{\sqrt{\tau_m}}{4}$, linking it to the membrane time constant. This scaling increases sensitivity to input for lower membrane time constants, while moderating capacitance growth for larger τ_m. This results in a gradient where integration speed and input sensitivity co-vary with τ_m, enabling a systematic investigation of how heterogeneity influences spiking dynamics under shared input conditions. Therefore, the heterogeneous gradient should be interpreted as a combined τ_m and C_m gradient rather than as an isolated variation of τ_m. This distinction is important when interpreting the results.

The membrane capacitance was scaled as a function of the membrane time constant to create a heterogeneous population with different temporal response properties. Larger values of τ_m were associated with larger capacitance values, resulting in slower voltage changes and longer temporal integration. Smaller values of τ_m were associated with lower capacitance values, making these neurons more sensitive to brief, temporally concentrated inputs. Thus, the gradient should be interpreted as a combined τ_m and C_m gradient, designed to produce systematic differences in how neurons respond to different temporal

input structures.

The spiking threshold was selected separately for experiments requiring spiking activity to induce an appropriate level of neural activity in the network. For experiments in which no spiking was desired, the threshold was set to 10^9 mV to prevent activity from occurring. Furthermore, to imitate realistic initial conditions, initial membrane potentials were sampled from a normal distribution with $\mu = -70.0$ mV and $\sigma = 5.0$ mV.

For additional parameter values, refer to Table 3.2.1.

Table 3.2.1: Additional Neuron and Synaptic Parameters

Parameter	Value	Description
E_L	-70 mV	Resting membrane potential
V_{reset}	-70 mV	Reset potential of the membrane
t_{ref}	2 ms	Duration of refractory period
$\tau_{syn,ex}$	2 ms	Excitatory synaptic decay time constant
$\tau_{syn,in}$	2 ms	Inhibitory synaptic decay time constant
I_e	0 pA	Constant input current

3.2.3 Parameter Validation

This validation step was performed to ensure that the neuron model behaved as expected with our chosen parameters before constructing the entire network.

The neuron model was first validated in isolation using a two-neuron setup without synaptic connections between them. One neuron was assigned a short membrane time constant, 5 ms (”fast” neuron), and the other a long membrane time constant, 80 ms (”slow” neuron). Three input arrangements were applied to both neurons: a single presynaptic spike, a brief spike burst, and regularly timed spiking at fixed intervals. The membrane potentials of both neurons were recorded using a built-in NEST voltmeter, with the expected result that the fast neuron would have a higher peak V under high-frequency input and the slow neuron under low-frequency input, confirming correct parametrisation of the neurons.

The validation was subsequently extended to a population of 76 uncoupled neurons with membrane time constants ranging from

5 to 80 ms. This broader test was designed to verify that the introduction of a continuous gradient of membrane time constants would not alter the assumed relationship between this parameter and membrane potential responses across different input frequencies. This experiment was replicated for four different input frequencies, from bursty to continuous. For each input, the membrane potential was recorded across the entire population to allow comparisons between the input types.

3.3 Input Design

To ensure comparable input conditions across experiments while preserving realistic variability, the inputs were normalised and supplemented with temporal jitter and stochastic spike transmission. The input-design experiments were simulated for 2000 ms in order to cover the full set of burst arrangements, including the most distributed 100-burst condition.

3.3.1 Energy-Based Normalisation

Input normalisation was applied to decrease the variations in spiking activity and network dynamics that arose from differences in input magnitude, and instead focus on variations arising from differences in the input temporal structure.

Input arrangements can vary in the number of bursts n_b, as well as in the number of spikes per burst n_s. To allow a fairer comparison across different arrangements, the inputs were normalised so that each arrangement delivered a similar amount of total energy.

The energy of a spike signal can be approximated by analogy with signal energy, $E \propto \int |x(t)|^2 dt$: the amplitude of a burst scales with the number of spikes it contains n_s, and the total energy it accumulates over n_b bursts, giving:

$$E \propto n_s^2 \times n_b$$

To keep E constant across arrangements, n_s must be scaled to

$$n_s \propto \frac{1}{\sqrt{n_b}}$$

This normalisation does not keep the total number of generated spikes constant. Instead, it keeps the equivalent signal-energy constant. This was chosen because the experiments compare temporally concentrated and temporally distributed inputs, and the instantaneous amplitude of a burst is expected to influence whether neurons cross the threshold. Thus, the normalisation preserves comparable input strength while allowing the temporal structure of the input to vary.

In practice, a reference arrangement was chosen to fix the overall scale, and the spikes per burst, or *activity*, for any other arrangement were derived from it:

$$n_s = n_{s,ref}\sqrt{\frac{n_{b,ref}}{n_b}} \tag{3.2}$$

where $n_{s,ref}$ and n_{b_ref} are the spikes per burst and number of bursts in the reference condition. This ensures that $n_s^2 \times n_b$ remains constant regardless of how the bursts are distributed.

By normalising input, variety in spiking activity and network dynamics arising from differences in input amplitude are decreased.

Two tests compared normalised with non-normalised input. Both of these tests had a simulation time of 1000 ms. The number of bursts in both tests were set to $[100, 50, 25, 10]$, representing regular to bursty input.

3.3.2 Peak membrane potential under normalised input

To further evaluate how the normalised burst conditions affected neurons across the heterogeneous gradient, the peak membrane potential was measured for each value of τ_m. For each burst condition, the maximum membrane potential reached during the simulation was

recorded for each neuron. This made it possible to compare how close different parts of the τ_m gradient came to the firing threshold under the same normalisation rule.

This analysis was performed without recurrent network interactions, so that the measured peak potentials reflected the direct effect of the external input and the τ_m gradient rather than excitation or inhibition from other neurons. The peak potential for neuron i was defined as

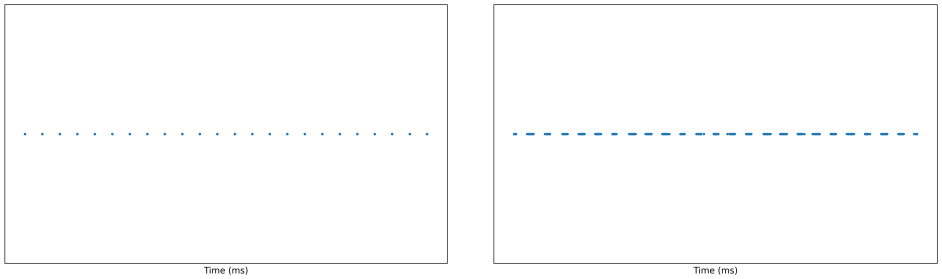
$$V_{\text{peak},i} = \max_{t \in [0,T]} V_i(t)$$

where $V_i(t)$ is the membrane potential of neuron i at time t, and T is the total simulation duration. The resulting peak-potential curves were used to evaluate whether the normalised inputs produced similar threshold proximity across burst conditions.

3.3.3 Input Variability and Stochastic Transmission

The `pulsepacket_generator` produces spike trains based on a sequence of pulse times. For each pulse time, it emits a specified number of spikes (*activity*), with normally distributed temporal jitter around the pulse centre, controlled by the standard deviation σ_{gen} [30]. A higher standard deviation results in a more temporally dispersed input, whereas lower values produce more temporally precise and bursty input. Fig. 3.3.1 illustrates how adjusting the standard deviation of the `pulsepacket_generator` affect its output. The `pulsepacket_generator` was chosen over alternatives such as the `spike_generator` because it enables independent control of spike count per burst, temporal precision, and spike-time variability.

The `pulsepacket_generator` was connected to a `parrot_neuron`, which replicates incoming spikes to all neurons in the network. The outgoing connections used `bernoulli_synapse` synapses, where each spike was transmitted independently for each neuron with probability p_{tx}. This introduced stochastic variability in spike transmission across neurons. When probabilistic transmission was introduced, the number



$\sigma_{\text{gen}} = 0$ ms $\sigma_{\text{gen}} = 4$ ms

Figure 3.3.1: Example spike trains generated by the `pulsepacket_generator` for two temporal-jitter values. Both panels use the same burst times and total spike count, but σ = 0 ms produces synchronous spike packets, whereas σ = 4 ms spreads spikes around each packet centre.

of generated spikes was scaled by a factor of $\frac{1}{p_{\text{tx}}}$. This compensated for the expected spike loss caused by stochastic transmission, thereby preserving approximately the same expected number of received input spikes per neuron.

A `poisson_generator` was connected to all neurons to provide stochastic background activity. Each neuron receives an independent input as noise. By connecting the `poisson_generator`, stochastic synaptic-like input introduces an approximation of ongoing background activity, which can be observed in biological neural systems [31]. The firing rate of the `poisson_generator` was set to $\lambda = 250$ Hz over a simulation interval of 2000 ms, corresponding to an expected spike count of approximately 500 spikes per neuron during each simulation. When background noise was introduced using the `poisson_generator`, the generator weight was reduced from 0.20 to 0.17 in order to approximately preserve the overall level of synaptic input to the network. This adjustment was made to isolate the behavioural effects of stochastic background activity while minimising effects arising from changes in the total number of input spikes.

The simulations were designed to evaluate how temporal jitter, stochastic spike transmission, and background noise influence neuronal responses and network activity. In particular, the tests investigated how neurons with different membrane time constants respond to temporally concentrated versus temporally dispersed inputs under varying

levels of stochasticity.

In all tests, four normalised input configurations were designed: 100 total bursts, 50 total bursts, 25 total bursts, and 10 total bursts. The bursts were distributed evenly in each simulation. The total number of spikes for each configuration was normalised according to Eq. 3.2

3.4 Network Architecture

Given the defined neuron dynamics and standardised input, the network architecture determines how activity is distributed and propagated through its spatial embedding, synaptic connectivity and excitation-inhibition balance.

3.4.1 Network Inhibition

To balance excitatory activity within the network and prevent excitation from dominating spiking activity, inhibitory neurons were introduced. The total number of neurons in the network is defined as

$$N = N_e + N_i$$

where N denotes the total number of neurons, N_e the number of excitatory neurons, and N_i the number of inhibitory neurons. An inhibitory-to-excitatory ratio of 1:4 was used, corresponding to approximately 20% inhibitory neurons, consistent with biological cortical networks and widely used in computational models [20].

Neurons were designated as inhibitory by assigning a negative value to their outgoing synaptic weights. Consequently, spikes emitted by inhibitory neurons reduced the membrane potential of their postsynaptic targets, thus counteracting excessive excitation and contributing to stable network dynamics. The inhibitory weight parameter effectively determines the overall inhibitory strength in the network, and is therefore the primary tool used to regulate the balance between excitation and inhibition. Appropriate tuning of the inhibitory weight is essential to achieve stable network dynamics, as overly weak inhibition can

lead to excessive synchronous firing, while overly strong inhibition can suppress network activity [7].

3.4.2 Spatial Embedding

Neurons were placed on a one-dimensional axis, with neurons with similar membrane time constant values placed close to each other. To avoid boundary effects arising from edge neurons, the network was implemented with periodic boundary conditions (edge wrapping). In this configuration, the first and last neurons are considered adjacent in space, forming a circular topology.

To ensure consistency with this topology, the gradient of the membrane time constant was also defined periodically. Specifically, the membrane time constant increases from a minimum value τ_{min} to a maximum value τ_{max} toward the centre of the network, and then decreases symmetrically. As a result, the first neuron has τ_{min}, the central neuron has τ_{max}, and the last neuron has the same membrane time constant as the second neuron, ensuring continuity at the boundary. This can be seen in Figure 3.4.1

3.4.3 Network Connectivity

Inter-neuron connections were formed using the NEST built-in `pairwise_bernoulli` connection-rule. The `pairwise_bernoulli` has the probability p to connect a source S to a target, T. Note that this is not symmetric relationship by default. p in this case is the NEST built-in parameter `spatial_distributions.gaussian()`, which normally distributes connections based on their spatial distance. Connection probabilities depended on the periodic distance d between source and target neurons. For each ordered source-target pair, a connection has a probability of forming of

$$p(d) = e^{\frac{-d^2}{2\sigma^2}} \tag{3.3}$$

For excitatory neurons, the standard deviation σ was 2 spatial units and for inhibitory 8 spatial units. Self-connections were not allowed and

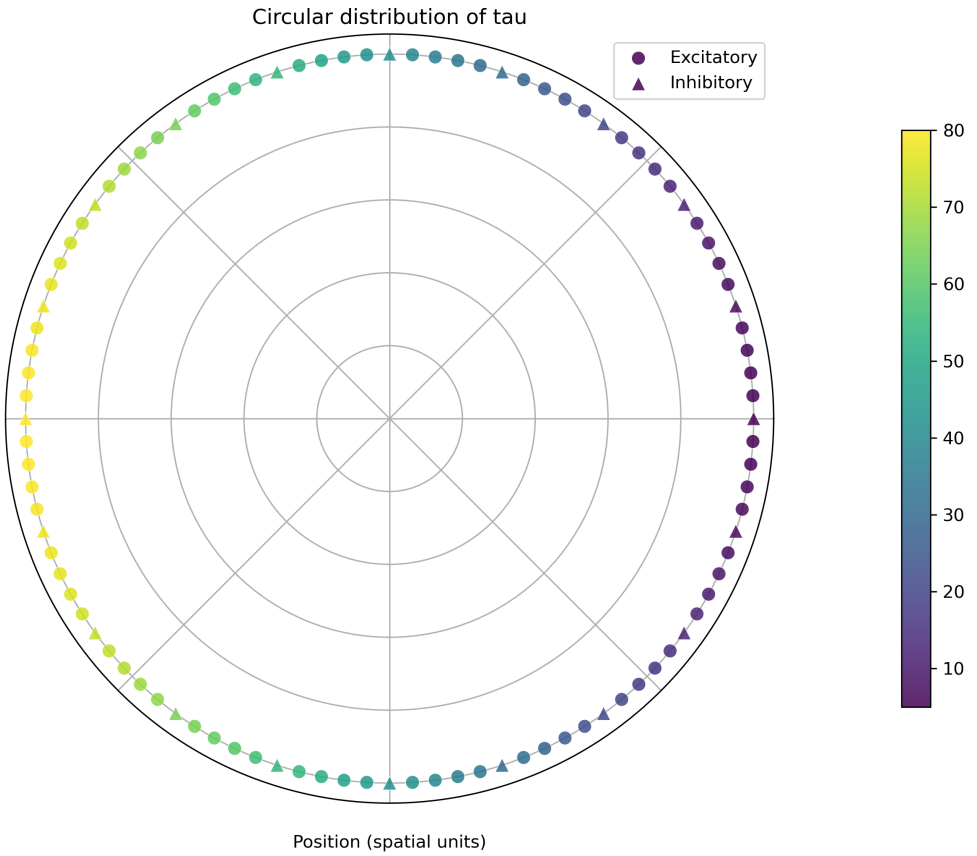


Figure 3.4.1: Periodic spatial embedding of the heterogeneous neuron population. The circular layout removes boundary effects by making the first and last neurons adjacent, while colour indicates τ_m and marker shape indicates excitatory or inhibitory neuron identity.

multiple connections between the same source-target pair were also not allowed.

All neurons were connected to the `pulsepacket_generator`, via `parrot_neuron`, with a synaptic weight of 0.17. The synaptic weight from excitatory neurons was set to 4.0. The synaptic weight from inhibitory neurons was set to -32.0.

The delay for all connections was kept at a default 1 ms.

The intended connectivity structure consisted of short-range excitatory connections and long-range inhibitory connections. To verify this connectivity structure, all inter-neuron connections were classified into one of four categories according to presynaptic and postsynaptic neuron types:

- $E \to E$ (Excitatory to excitatory)
- $E \to I$ (Excitatory to inhibitory)
- $I \to E$ (Inhibitory to excitatory)
- $I \to I$ (Inhibitory to inhibitory)

The connection counts for each category were extracted from the network and visualised using a connectivity matrix as seen in Figure 3.4.2. This analysis was used to assess whether connectivity patterns were proportionally distributed according to the spatial distance between neurons.

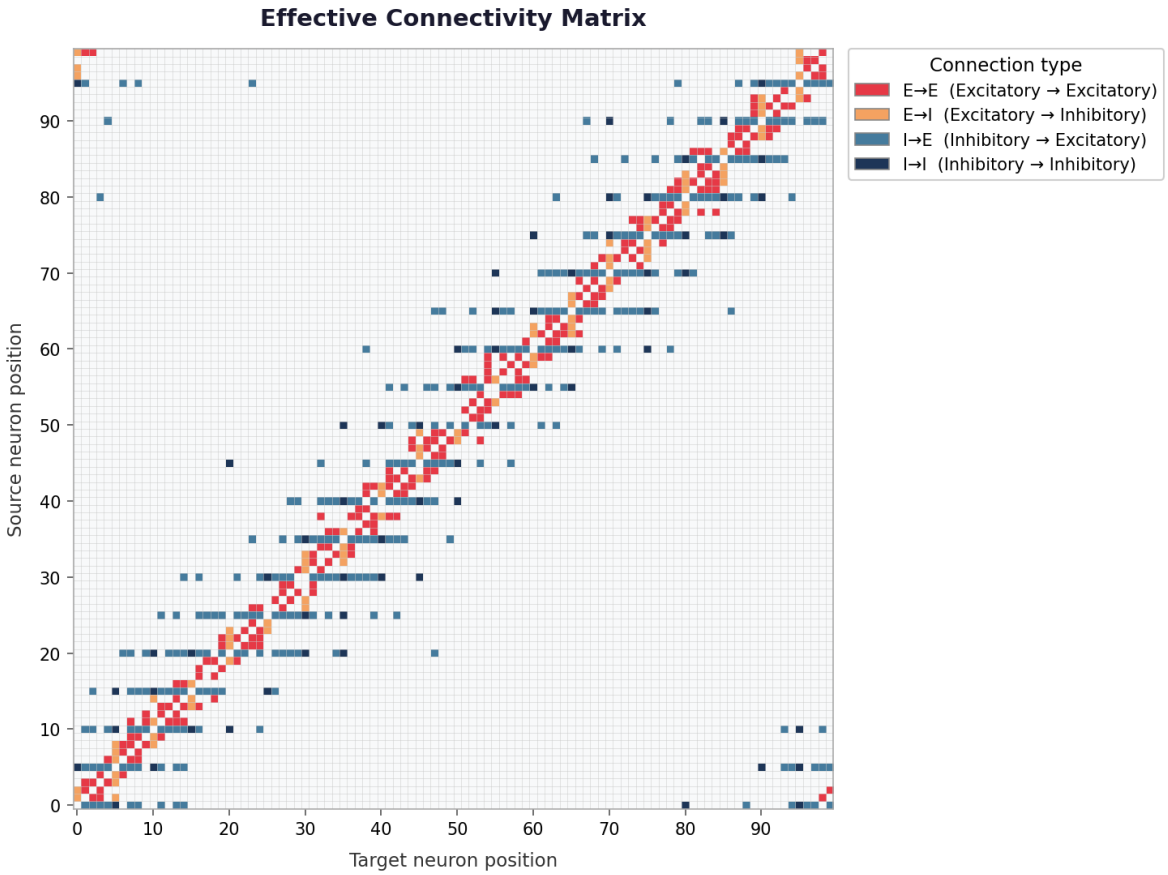


Figure 3.4.2: Distance-dependent connectivity in the recurrent network. Each point marks a directed synaptic connection between a source and target neuron, separated by connection type. The dense diagonal structure reflects short-range connectivity, while broader off-diagonal patterns show longer-range inhibitory connections.

3.4.4 Comparing External and Recurrent Activity

To verify that the network maintained a balanced ratio between activity generated by the external generator and by the network's own neurons, measurements of both external and internal drive were conducted. The external drive originated from the `pulsepacket_generator`, while the internal drive was generated by spikes emitted from excitatory neurons and regulated by inhibitory neurons. By adjusting the synaptic weights between the generator and the neurons, as well as the connections between internal neurons, different ratios between external and internal drive could be obtained and evaluated.

Three different network configurations were analysed: a network without synaptic connections, a connected network containing only excitatory neurons, and a connected network containing both excitatory and inhibitory neurons. In the configurations without inhibitory synaptic connections, the inhibitory synaptic weight was set to 0, such that neurons designated as inhibitory in the model effectively functioned as neurons without outgoing connections. In each configuration, the individual driving factors were measured separately, making it possible to evaluate the relationship between external and internal drive within the network. All networks used $N = 100$ and two input arrangements of 10 and 100 bursts each using $n_{s,ref} = 10000$ with $n_{b,ref} = 100$. The generator weight was set to a factor of 0.17. The spiking threshold V_{th} was set at -55 mV. For each network configuration and burst time, the membrane potential, spike times, and spiking activity were measured and plotted alongside the comparison between the external and internal activity.

3.5 Connected Heterogeneous Network

The connected heterogeneous network was the primary experimental model used to evaluate the effect of continuously varying neural heterogeneity on spiking activity in a one-dimensional spatial network. Unlike previous validation experiments that isolated a single property of the network, this network incorporated full recurrent connectivity, spatial embedding, and excitation-inhibition balance, thereby enabling investigation of how intrinsic neuronal heterogeneity interacts with structured network dynamics.

The network consisted of $N = 1000$ neurons distributed along the previously defined one-dimensional periodic spatial axis. Each neuron was assigned a membrane time constant according to the continuous gradient described in Section 3.2.2, ranging from $\tau_{\text{min}} = 5$ ms to $\tau_{\text{max}} = 80$ ms. Excitatory and inhibitory neurons were distributed throughout the gradient while preserving the global inhibitory ratio of approximately 20%.

Inter-network connectivity was implemented using the Gaussian distance-dependent connectivity model described in Section 3.4.3.

For each experimental condition, neuronal activity was quantified using *firing rates* and *spike timing*. These measurements allowed analysis of both individual neuronal responses and emergent population-level dynamics across the membrane time constant gradient.

For the connected heterogeneous-network simulations, the network was evaluated under the three input-strength regimes described in Section 3.5.1. Each regime was tested using the same four burst conditions: 10, 25, 50, and 100 bursts. The purpose of this comparison was to evaluate whether the relationship between membrane time constant and firing rate was robust to input scaling, or whether the observed activity patterns depended on the absolute strength of the external drive.

3.5.1 Input-Strength Regimes

Although input normalisation was used to make the burst conditions comparable, preliminary simulations showed that the activity of the connected heterogeneous network was highly sensitive to the absolute input strength. Therefore, in addition to the normalised input structure, three input-strength regimes were evaluated in the main heterogeneous-network simulations.

The first regime was the normalised baseline input. In this regime, all burst conditions were normalised according to Eq. 3.2 and shared the same global input strength. This regime was treated as the main comparable condition, since differences between burst conditions were primarily due to temporal structure rather than differences in overall input magnitude. The generator weight used was 0.17.

The second regime used increased global input strength. The same normalisation rule was preserved, but the overall input gain was increased equally for all burst conditions. This regime was used to test whether the qualitative firing-rate trends observed in the baseline condition were robust to changes in input magnitude. The generator weight used was 0.20.

The third regime used condition-specific input rescaling. In this regime, the input strength of the intermediate burst conditions was increased in order to test whether activity could be recovered for 25 and 50 bursts. This regime was treated as exploratory, because it relaxed the normalisation constraint between burst conditions. As a result, this regime was not used as the main comparable result, but rather as a calibration experiment showing that intermediate activity could be induced by changing the relative input strength. The generator weights used were 0.17 for the edge burst conditions and 0.20 for the intermediate burst conditions. Table 3.5.1 summarises the input scaling regimes.

Together, these regimes were used to distinguish between effects caused by temporal input structure and effects caused by input scaling. This distinction is important because the aim of the normalisation procedure was to reduce differences in input magnitude while preserving differences in burst timing.

Regime	**Normalisation preserved**	**Gen. weight**	**Input scaling**	**Purpose**
Baseline normalised input	Yes	0.17	Uniform generator weight across burst conditions	Main comparable condition
Increased global input	Yes	0.20	Increased uniform generator weight across burst conditions	Sensitivity test
Condition-specific rescaling	No	0.17 and 0.20	Increased gain for intermediate burst conditions	Exploratory recovery of intermediate activity

Table 3.5.1: Summary of input scaling regimes.

The normalised baseline regime was used as the primary comparable condition for evaluating how temporal burst structure influenced activity across the membrane time constant gradient. The additional input regimes were included to determine whether the observed firing-rate trends were robust to changes in input strength, and

whether reduced activity in the intermediate burst conditions could be recovered through selective rescaling.

This design allowed the main heterogeneous-network experiment to assess not only whether different burst structures preferentially activated different regions of the τ_m gradient, but also whether these effects were stable under changes in input scaling.

3.6 Analysis metrics

Spiking activity was quantified using firing rate and spike timing across the membrane-time-constant gradient. The firing rate of neuron i was computed as:

$$r_i = \frac{n_i}{T}$$

Where n_i is the number of spikes emitted by neuron i during the analysis interval T. Population firing-rate visualisations were obtained by plotting r_i as a function of τ_m. Spike timing was analysed using raster plots in which spike times were grouped by τ_m. For the connected heterogeneous network experiment, firing-rate trends were visualised by smoothing r_i along the ordered τ_m gradient using a moving-average filter. The smoothing window was set to 12% of the number of neurons, with a minimum window size of three neurons. Edge padding was applied before filtering to avoid artificial boundary effects. The smoothed curves were used only for visualisation, while the individual firing rates r_i were plotted as low-opacity points to show the underlying variability.

To compare external and internal activity, external drive was defined as the average number of spikes delivered from the `pulsepacket_generator` to network neurons, while internal activity was defined as spikes received by recurrently connected network neurons. The relationship between these sources was quantified using weighted spike counts by synaptic input.

Chapter 4

Results

This chapter presents the results in four stages. First, the selected neuron parametrisation is validated in isolated and uncoupled populations. Second, the effects of input normalisation, temporal structure, jitter, stochastic transmission, and background noise are evaluated. Third, recurrent network configurations are compared in order to separate externally driven activity from internally generated recurrent activity. Finally, the full connected heterogeneous network is analysed across three input-strength regimes to determine how spiking activity varies across the membrane-time-constant gradient, and whether these activity patterns are robust to input scaling.

4.1 Neuron Validation

4.1.1 Parameter validation

The two-neuron validation confirmed that the `iaf_psc_exp` model responded as expected under all three input arrangements. Figure 4.1.1 shows the recorded membrane potentials for the fast neuron ($\tau_m = 5$ ms) and the slow neuron ($\tau_m = 80$ ms).

Under a single presynaptic spike, the fast neuron exhibited a sharper but higher peak, while the slow neuron showed a lower amplitude with a longer tail. During the brief spike burst, the fast neuron accumulated

charge more rapidly and reached the highest peak. Under regular input, the slow neuron reached higher peak membrane potentials, confirming that the parametrisation was favourable for studying neural heterogeneity, which is consistent with expected results from scaling $C_m = \sqrt{\tau_m}/4$, which co-varies input sensitivity with integration speed.

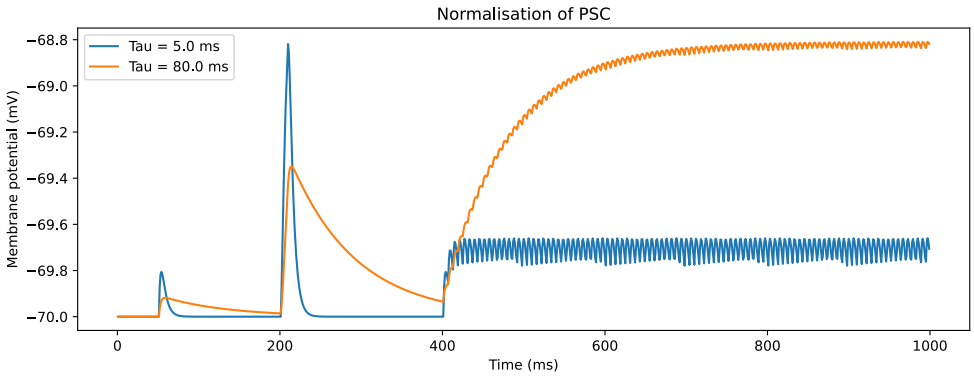


Figure 4.1.1: Membrane potential traces of the fast ($\tau_m = 5$ ms) and slow ($\tau_m = 80$ ms) neurons under three input configurations.

4.1.2 Population Validation Across Input Frequencies

Figure 4.1.2 summarises the membrane potential across the population of 76 uncoupled neurons ($\tau_m \in [5, 80]$ ms) for all four input frequencies.

For bursty input, the peak membrane potential decreased monotonically with increasing τ_m, and the reverse behaviour can be spotted for regular input. The introduction of a continuous τ_m gradient did not disturb the expected monotonic relationship between time constant and potential response.

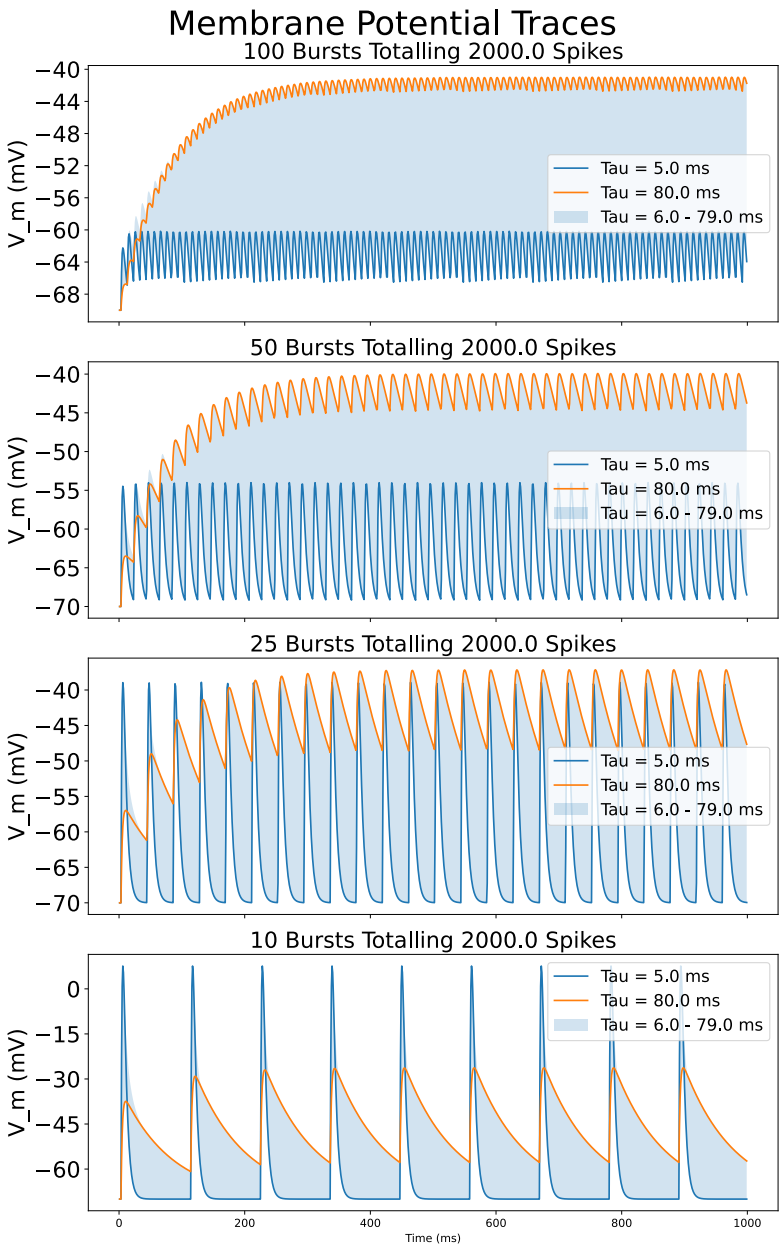


Figure 4.1.2: Membrane-potential responses of uncoupled neurons with τ_m values spanning 5-80 ms under four temporal input arrangements. Bursty input produces larger peak responses in low-τ_m neurons, whereas more regular input favours high-τ_m neurons , confirming that the continuous τ_m gradient preserves the expected temporal-integration pattern.

4.2 Effect of Input Design Before Recurrent Connectivity

These experiments evaluated how temporal structure, jitter, stochastic transmission, and background noise changed which parts of the heterogeneous neuron population were most strongly activated.

4.2.1 Effect of Normalisation

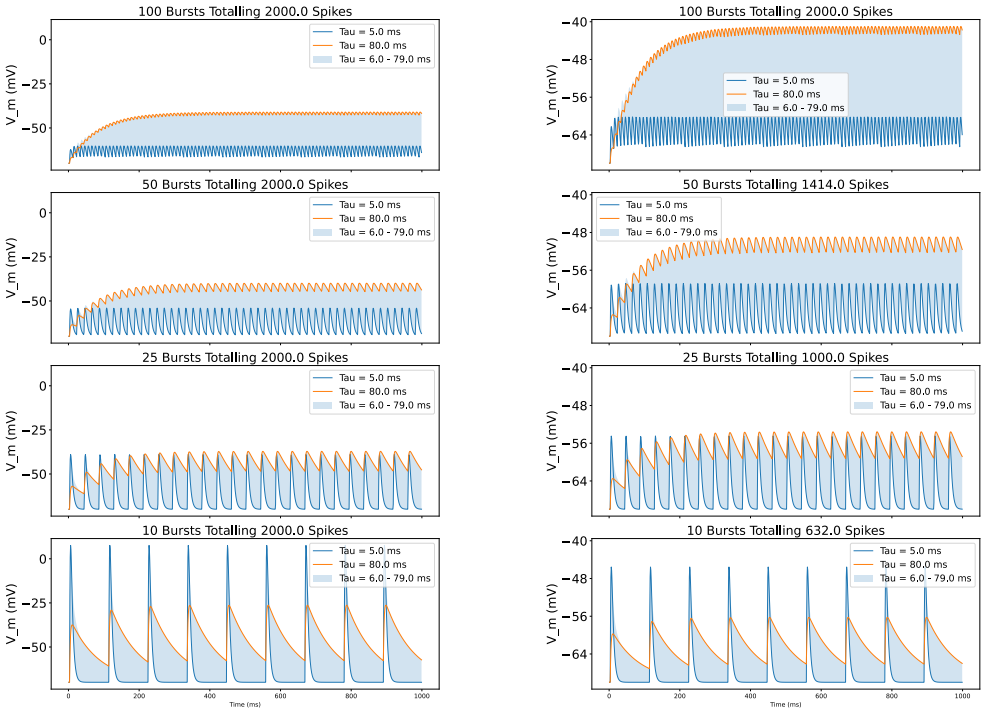
Normalising the input results in more comparable peak membrane potentials across different input configurations, which in turn facilitates consistent selection of the threshold V_{th}. Fig. 4.2.1 shows the membrane potential traces of two identical networks under different input arrangements. After normalisation, shown in Fig. 4.2.1b, the peak membrane potentials become more similar across conditions compared to pre-normalisation, shown in Fig. 4.2.1a, while the overall structure of each neuron's membrane potential trace is preserved.

However, as shown in Fig. 4.2.1b, the peak membrane potentials are not identical even after normalisation.

4.2.2 Peak membrane potential under normalised input

To further examine the effect of normalisation, the peak membrane potential was measured across the τ_m gradient for each burst condition. This analysis was performed to evaluate whether the normalised inputs brought different parts of the heterogeneous population equally close to the firing threshold.

As shown in Fig. 4.2.2, the peak membrane potential was not equal across burst conditions after normalisation. Instead, the largest peak potentials occurred for the temporal edge conditions. The 10-burst condition produced the highest peak potentials for neurons with low τ_m, while the 100-burst condition produced the highest peak potentials for neurons with high τ_m. In contrast, the intermediate 25- and 50-burst conditions remained lower across much of the gradient.



(a) Non-normalised input (b) Normalised input

Figure 4.2.1: Effect of energy-based input normalisation on membrane-potential traces across four burst conditions. Non-normalised input produces large differences in peak voltage between burst conditions, while normalised input reduces these differences while preserving the temporal structure of each condition.

This shows that the energy-based normalisation reduced differences in overall input magnitude, but did not make all burst conditions equally effective at driving membrane potentials toward threshold. The temporal structure of the input still interacted with the heterogeneous τ_m gradient. Temporally concentrated input produced stronger peak responses in the low-τ_m region, whereas temporally distributed input produced stronger peak responses in the high-τ_m region.

The intermediate burst conditions did not produce the highest peak response in either end of the gradient. This suggests that, under the chosen normalisation rule, these inputs occupy an intermediate temporal regime that is less effective at maximising the membrane response of either fast-responsive or slow-integrating neurons.

Therefore, the normalisation procedure preserved comparable input energy, but the resulting membrane-potential responses were still strongly shaped by temporal structure and neuronal heterogeneity.

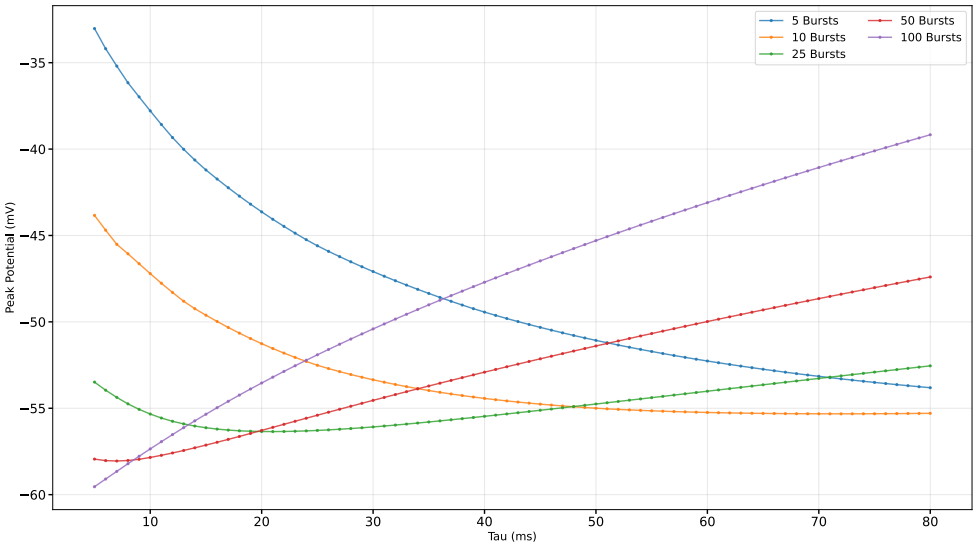


Figure 4.2.2: Peak membrane potential across the τ_m gradient for each normalised burst condition. The 10-burst condition produces the highest peak potentials for low-τ_m neurons, while the 100-burst condition produces the highest peak potentials for high-τ_m neurons. The intermediate 25- and 50-burst conditions remain lower across much of the gradient, showing that normalised input energy does not produce equal threshold proximity across burst conditions.

4.2.3 Effect of Temporal Jitter and Probabilistic Transmission

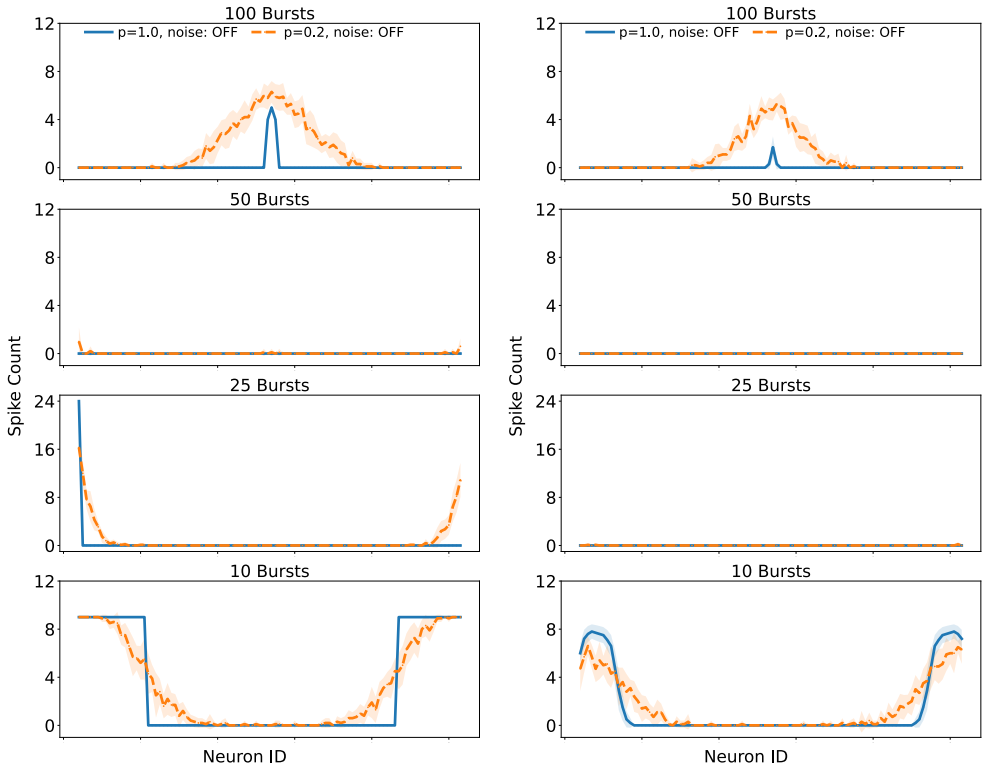
Increasing the standard deviation corresponds to a greater temporal spread of the input, whereas a lower standard deviation produces a temporally concentrated input pattern. Decreasing the standard deviation increased the structure of the spiking activity in bursty input environments. Furthermore, comparing Fig. 4.2.3a with Fig. 4.2.3b show that neurons with low membrane time constants exhibited higher activity for $\sigma_{\text{gen}} = 0$ ms compared to $\sigma_{\text{gen}} = 4$ ms.

Introducing probabilistic transmission increased the variability of the input. This additional variability altered the spiking dynamics, as neurons that would otherwise remain below the firing threshold gained a probabilistic firing, while neurons that would otherwise fire occasionally failed to reach the threshold due to stochastic effects.

For 100 bursts, Fig. 4.2.3a and Fig. 4.2.3b show similar overall spiking activity, although the case with $p_{\text{tx}} = 0.2$ exhibited higher overall activity than $p_{\text{tx}} = 1.0$. For 25 bursts, the spiking activity approached zero as the standard deviation increased, as shown in Fig. 4.2.3. Furthermore, for 10 bursts, the activity was lower overall for inputs with $\sigma_{\text{gen}} = 4$ ms.

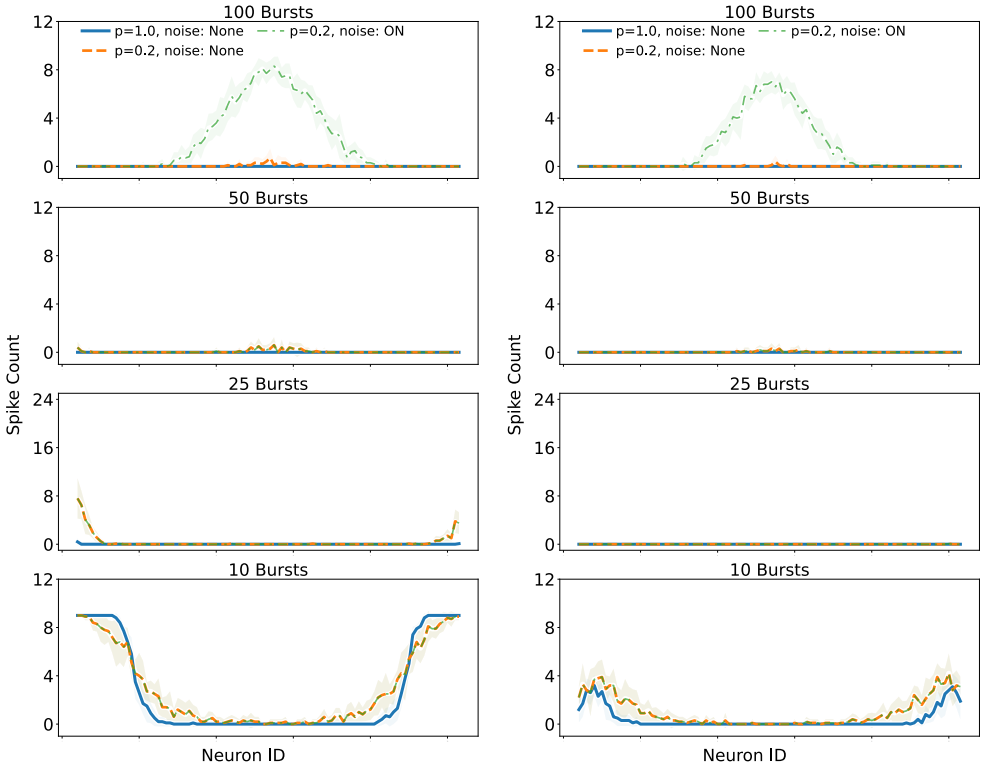
4.2.4 Effect of Background Noise

The background noise generated by the `poisson_generator` adds variability in spiking activity. For the 100-burst input configuration, background noise increased spiking activity among neurons with high membrane time constants. The spiking activity for neurons with low membrane time constant values did not change when background noise was introduced. This background activity appeared to produce effects similar to those observed with stochastic spike transmission by adding variability.



(a) $\sigma_{\text{gen}} = 0$ ms. (b) $\sigma_{\text{gen}} = 4$ ms.

Figure 4.2.3: Firing activity across neuron position for exact and probabilistic synaptic transmission without background noise. Panels compare $\sigma_{\text{gen}} = 0$ ms and $\sigma_{\text{gen}} = 4$ ms temporal jitter across the 100-, 50-, 25-, and 10-burst conditions. Probabilistic transmission increases response variability, while temporal jitter weakens burst-driven firing.



$\sigma_{\text{gen}} = 0$ ms. $\sigma_{\text{gen}} = 4$ ms.

Figure 4.2.4: Firing activity across neuron position for exact and probabilistic synaptic transmission with background noise enabled. Panels compare $\sigma_{\text{gen}} = 0$ ms and $\sigma_{\text{gen}} = 4$ ms temporal jitter across the 100-, 50-, 25-, and 10-burst conditions. The added Poisson background increases activity across burst conditions and makes the responses for exact and stochastic transmission more similar.

4.3 Recurrent Network Control

For both burst protocols, all neurons received consistent external input from the generator, with each neuron receiving approximately equal external stimulation. In the 100-burst configuration, external input was distributed continuously across the simulation, producing a relatively constant drive, while for the 10-burst configuration, the input was much more concentrated around the bursts, increasing instantaneous external drive.

For all three network and burst conditions, Figures 4.3.1, 4.3.2, and 4.3.4 contain activity plots which include a spike timing raster plot with burst timings marked by red vertical lines at the top of the figure, inhibitory spikes marked in red and excitatory spikes marked in blue. The activity plots also contain the firing activity for each excitatory and inhibitory neuron measured in spike counts. For conditions with network connections, a bar graph was also included in Figures 4.3.3b, 4.3.3a, 4.3.5b, 4.3.5a, which compares the weighted difference between external and internal input, with inhibitory neurons marked by red vertical lines.

Note that in networks without inhibitory synapses, the inhibitory synaptic weight was set to 0. Neurons designated as inhibitory prior to the introduction of inhibition therefore functioned as neurons without outgoing connections.

4.3.1 No Recurrent Connections

When synaptic connections were absent, neuron activity was determined only by external input and intrinsic neuronal parameters. This configuration therefore served as a baseline for identifying which parts of the τ_m gradient were directly activated by each input arrangement. Figure 4.3.1 shows neuron activity and membrane potential for the network without recurrent connections.

4.3.2 Excitatory Recurrent Network

The introduction of excitatory synapses led to activity levels exceeding the externally driven baseline. In particular, neurons with low

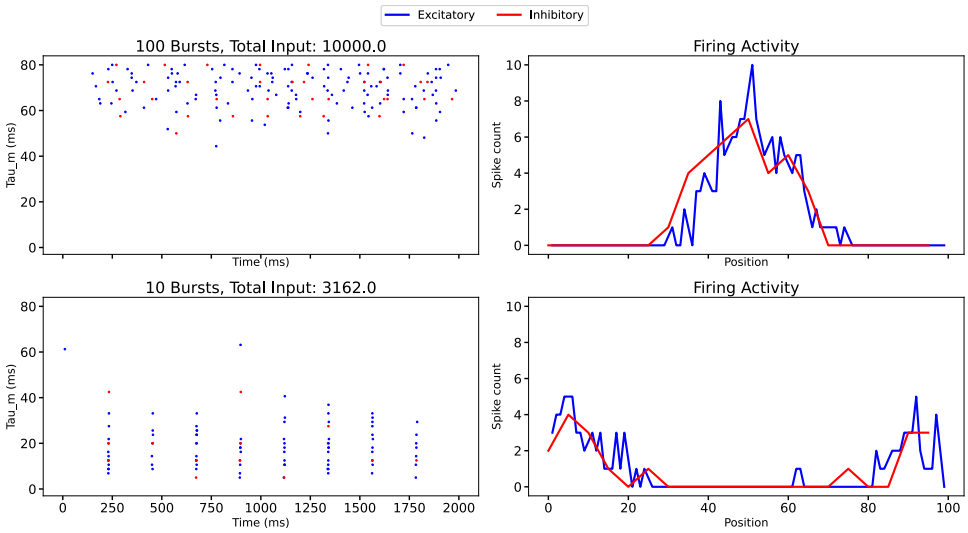


Figure 4.3.1: Activity in the network control condition without recurrent synaptic connections. Raster plots show spike timing across τ_m for 100- and 10-burst inputs, while spike-count panels show that activity is driven only by external input and intrinsic τ_m-dependent response properties.

membrane time constants exhibited a substantial increase in spiking activity, rising from a maximum of approximately 5 spikes in the non-connected network (Fig. 4.3.1) to around 400–500 spikes after the introduction of excitatory synapses (Fig. 4.3.2).

This increase is likely due to local recurrent excitation: neurons that are highly sensitive to external input tend to spike more readily, thereby providing additional excitatory input to nearby neurons with similar input sensitivity. This results in a self-amplifying activity pattern. The effect is especially pronounced in the region of the parameter space that is most responsive to the input condition.

The overall increase in spiking activity in the excitatory recurrent network, compared to the network without recurrent connections, is reflected in the recurrent drive shown in Fig. 4.3.3. As illustrated in Fig. 4.3.3b, the recurrent excitatory input is substantially higher than the external input in the 10-burst condition. In contrast, Fig. 4.3.3a shows a more distributed recurrent response for the 100-burst condition, highlighting differences in the balance between external and recurrent drive across input regimes.

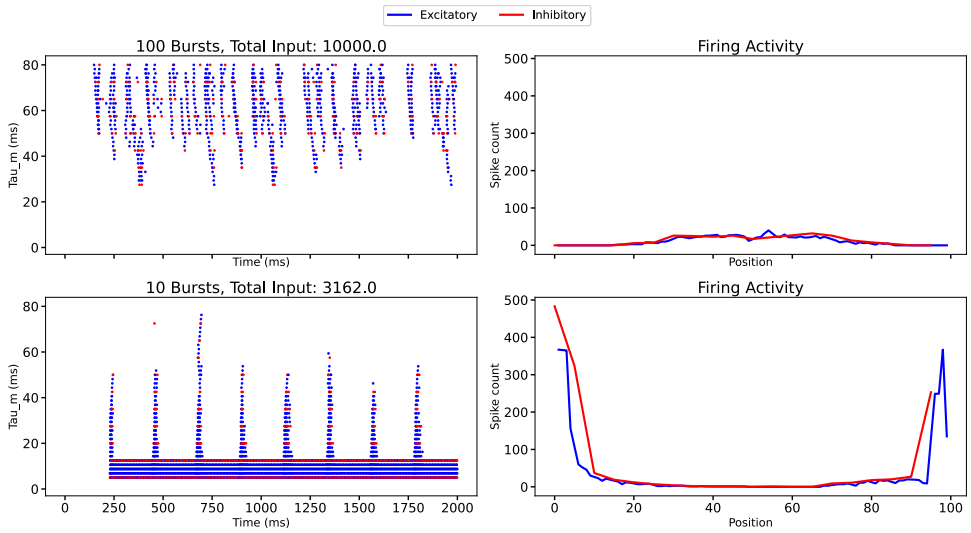
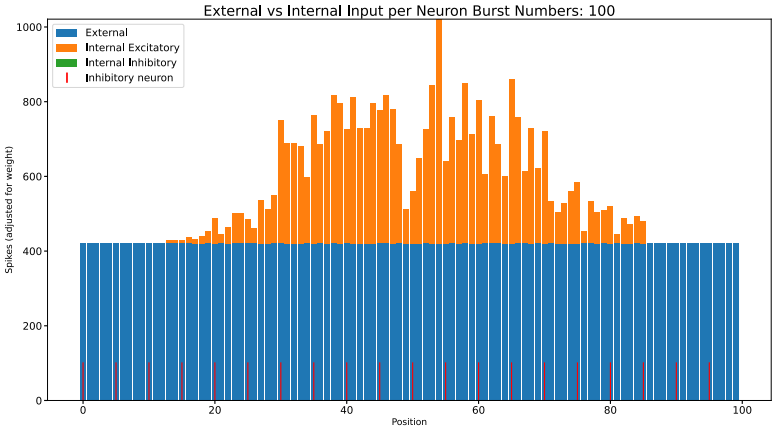


Figure 4.3.2: Activity in the recurrent network with excitatory synapses only. Compared with the disconnected control, recurrent excitation amplifies firing in neurons already recruited by the input, producing stronger activity for both the 100- and 10-burst conditions.

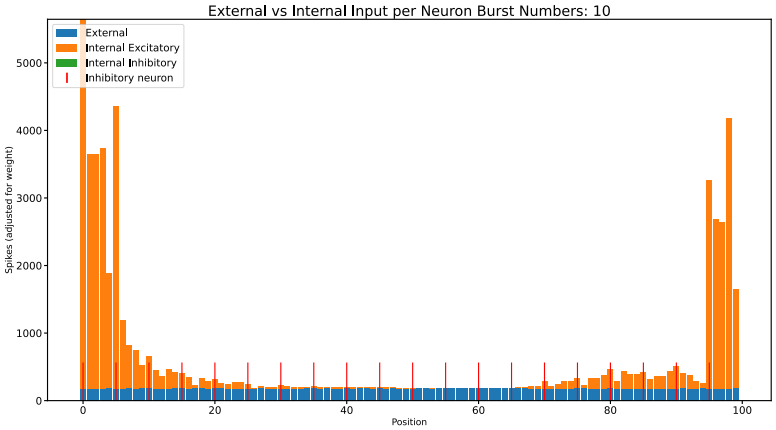
4.3.3 Effect of Inhibitory Regulation

The addition of inhibitory synapses altered the relationship between external input, recurrent excitation, and spiking activity. In the excitatory-only network, activity could be amplified through local recurrent connections, particularly in regions of the gradient that were already strongly driven by external input. When inhibitory neurons were included, this self-amplifying activity was constrained compared to the dynamics observed in Sec. 4.3.2.

As shown in Fig. 4.3.4, the excitatory–inhibitory network preserved the input-dependent structure observed in the network without recurrent connections (Fig. 4.3.1). Furthermore, the spiking activity of excitatory neurons appears to exhibit more pronounced dips in both burst conditions. This behaviour may be explained by the interaction between local excitation and inhibition. Neurons that fire early—including inhibitory neurons—can suppress neighbouring neurons that would otherwise have fired later. As a result, some neurons fail to spike during these inhibitory phases, which in turn reduces inhibitory feedback to the original firing neurons. This can allow the initially



(a) 100-burst condition



(b) 10-burst condition

Figure 4.3.3: Weighted external and recurrent input per neuron in the excitatory recurrent network for the 100-burst as well as 10-burst condition. (a): distributed input produces a broad internally generated excitatory response centred around the region that integrates regular input most strongly. (b): external drive is approximately uniform, while internally generated excitation is concentrated near the low-τ_m responsive regions activated by temporally bursty input.

active neurons to recover and fire again.

For the 10-burst condition, activity remained concentrated in the region of the network most responsive to temporally concentrated input. However, the inclusion of inhibition prevented uncontrolled amplification of activity. This is also reflected in Fig. 4.3.5b, where the recurrent inhibitory contribution increases in regions with strong excitatory drive. Inhibition therefore counteracted the strongest burst-driven responses and contributed to network stabilisation. Similarly, for the 100-burst condition, the recurrent inhibitory contribution increases in regions with strong excitatory drive, as shown in Fig. 4.3.5a.

Overall, the excitatory–inhibitory configuration provided a useful intermediate step before analysing the full heterogeneous network. It demonstrated that recurrent excitation can amplify input-dependent activity patterns, but that inhibition is necessary to prevent dominant runaway activity and to stabilise the network dynamics.

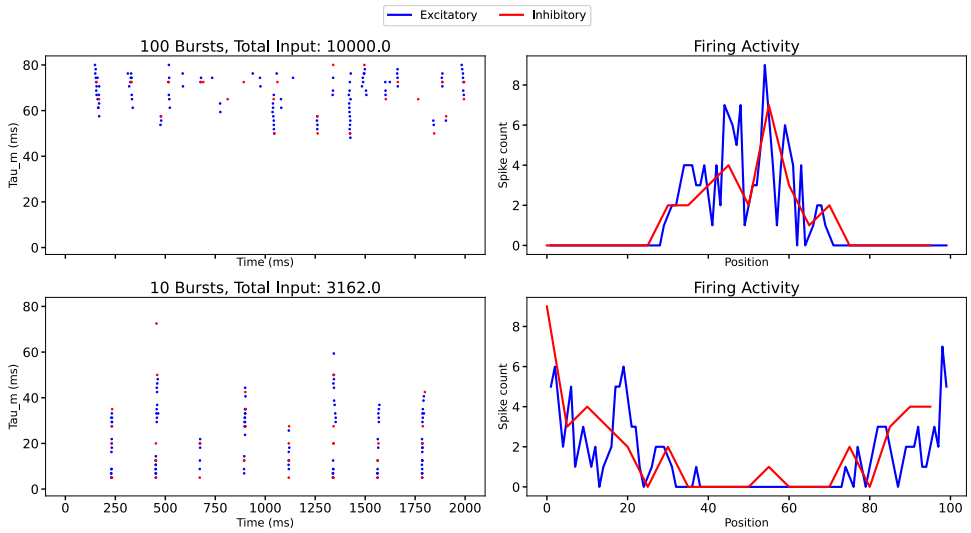
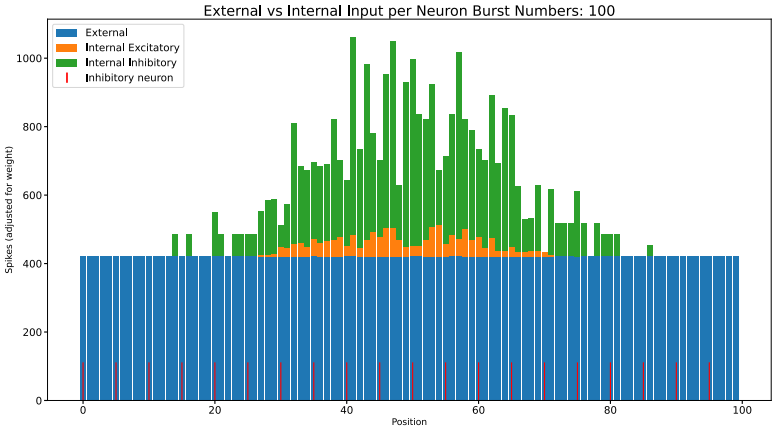
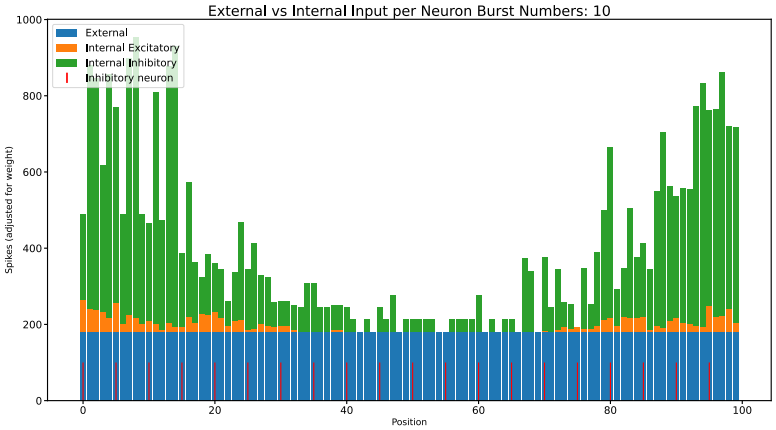


Figure 4.3.4: Activity in the connected excitatory-inhibitory recurrent network. Raster and spike-count panels compare 100- and 10-burst inputs, showing that inhibition limits the amplification observed in the excitatory-only network while preserving condition-specific recruitment across the τ_m gradient.



(a) 100-burst condition



(b) 10-burst condition

Figure 4.3.5: Weighted external, recurrent excitatory, and recurrent inhibitory input per neuron in the connected network for the 10-burst condition. (a): Distributed external input generates a broad recurrent response around intermediate-to-high τ_m positions, while inhibitory input limits excessive amplification. (b): Inhibition counteracts strong recurrent excitation in the burst-responsive low-τ_m regions, stabilising the burst-driven activity pattern.

4.4 Main Spatially Embedded Heterogeneous Network

The spatially embedded heterogeneous recurrent network was evaluated to determine how temporal input structure influenced spiking activity across the continuous membrane-time-constant gradient. In contrast to previous validation and input-design experiments, this network included the full spatial embedding, recurrent excitation, recurrent inhibition, and distance-dependent connectivity. Therefore, the firing-rate profiles reflected both single-neuron response properties and recurrent network interactions.

Across the simulations, the qualitative relationship between burst structure and firing-rate distribution depended strongly on input scaling. Under normalised baseline input, activity was clearest in the two temporal edge conditions. The 10-burst condition preferentially activated neurons with low membrane time constants, while the 100-burst condition preferentially activated neurons with high membrane time constants. In contrast, the intermediate 25- and 50-burst conditions produced weak activity across most of the τ_m gradient.

This result suggests that the network preserved temporal selectivity along the heterogeneous gradient, but only the most distinct temporal input structures produced clear firing under the baseline input strength. The sparse 10-burst input was most effective for fast-responsive low-τm neurons, whereas the distributed 100-burst input was most effective for slow-integrating high-τ_m neurons. The intermediate burst conditions appeared to fall between these response regimes and were not strong enough to reliably recruit either population.

4.4.1 Normalised Baseline Input

Under normalised baseline input, the highest activity occurred in the two temporal edge conditions. As shown in Fig. 4.4.1, the 10-burst condition produced the highest firing rates among neurons with low membrane time constants. Firing then decreased across the gradient, with little activity among neurons with larger τ_m. This indicates that temporally concentrated burst input was most effective at recruiting fast-responsive neurons.

The 100-burst condition showed the opposite trend. Activity was weak in the low-τ_m region and increased toward higher membrane time constants. This suggests that temporally distributed input was more effectively integrated by neurons with longer integration windows. Compared with the 10-burst condition, spiking in the 100-burst condition was less concentrated around individual burst events and more distributed across the simulation interval, as shown in Fig. 4.4.2.

In contrast, the intermediate 25- and 50-burst conditions produced weak activity under the same normalised input strength. These conditions did not strongly recruit either the low-τ_m or high-τ_m regions. This suggests that, under the chosen normalisation and threshold, the intermediate temporal structures fell between the two response regimes: they were not sufficiently concentrated to strongly activate fast-responsive neurons, and not sufficiently distributed to strongly activate slow-integrating neurons.

The baseline result therefore supports the existence of temporal selectivity across the heterogeneous gradient, but only for the most distinct temporal input structures. The weak activity in the intermediate conditions indicates that the expected progression from 10 to 100 bursts was not fully expressed under normalised input.

4.4.2 Increased Global Input Strength

To test whether the baseline firing-rate profiles were robust to input magnitude, the simulation was repeated with increased global input strength while preserving the same normalisation rule across burst conditions. In this regime, the generator weight was increased uniformly for all four burst conditions. Therefore, differences between burst conditions were still primarily determined by temporal structure, but the overall external drive was stronger.

Increasing the global input strength recruited more activity across the network, as shown in Fig. 4.4.4. In particular, the intermediate 25- and 50-burst conditions produced more visible firing than under baseline input. This indicates that their weak activity in the baseline regime was partly due to insufficient recruitment rather than a complete absence of responsiveness.

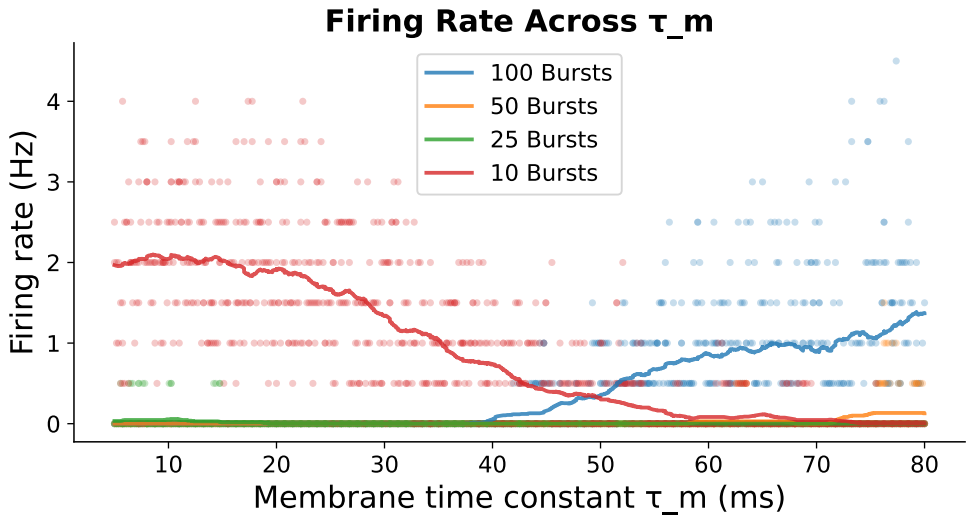


Figure 4.4.1: Firing rate across the τ_m gradient under normalised baseline input. Individual neurons are shown as low-opacity points, and smoothed trend lines show the average firing-rate profile for each burst condition. The 10-burst condition mainly recruits low-τ_m neurons, while the 100-burst condition mainly recruits high-τ_m neurons. The intermediate 25- and 50-burst conditions barely produce activity under the same normalised input strength.

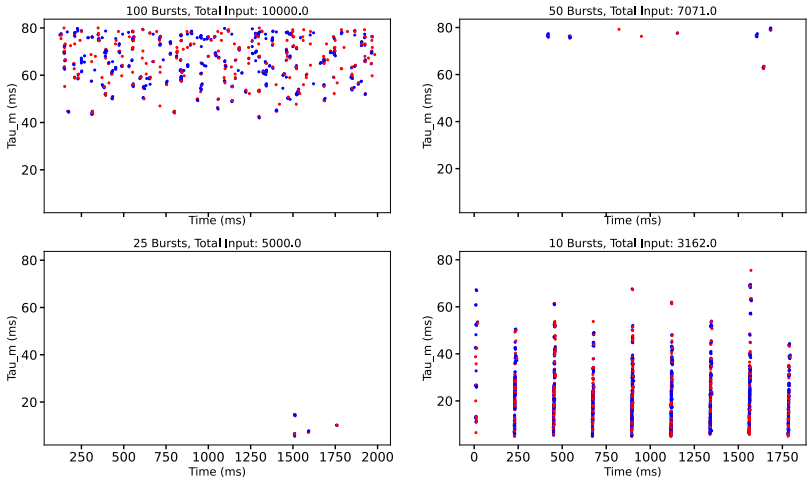


Figure 4.4.2: Spike timings under normalised baseline input. The 10-burst condition produces spikes concentrated around burst times and primarily among low-τ_m neurons. The 100-burst condition produces more distributed spiking shifted toward higher τ_m values. The intermediate burst conditions show limited spiking activity.

However, stronger input also changed the firing-rate profile of the 100-burst condition, see Fig. 4.4.3 Under baseline input, the 100-burst response increased most clearly toward larger τ_m values. Under increased global input, this monotonic trend was reduced, and the highest-τ_m region no longer showed the same simple increase in activity. This suggests that the firing-rate distribution was not determined only by the temporal integration properties of isolated neurons. Instead, recurrent excitation and inhibition also shaped the response.

One possible explanation is that stronger input allows more sensitive neurons to reach threshold earlier, which can then recruit recurrent inhibition before slow-integrating neurons reach their strongest response. This mechanism was not isolated directly in the present experiment, but the altered firing-rate profile shows that input scaling changes the balance between external drive and recurrent network interactions.

Thus, increasing the global input strength improved recruitment of the intermediate burst conditions, but also reduced the interpretability of

the firing-rate slope as a direct consequence of the τ_m/C_m gradient alone.

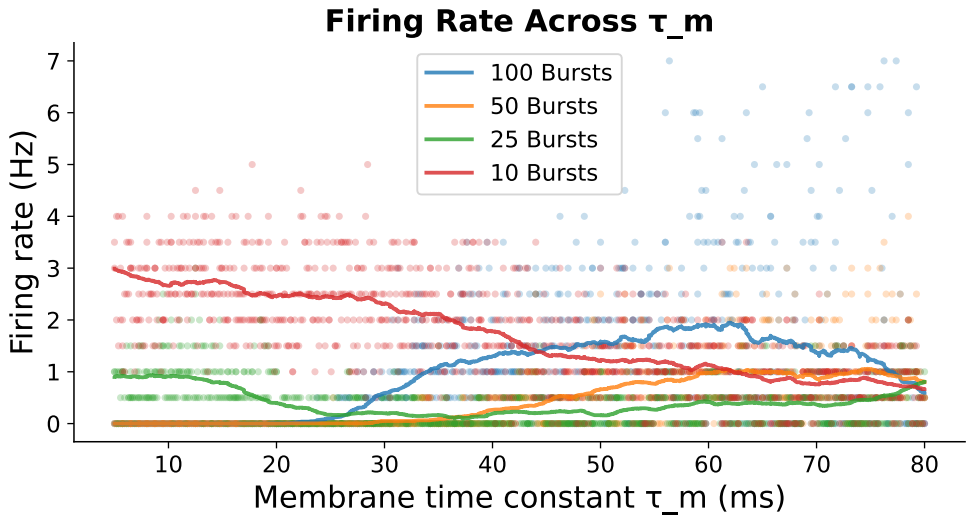


Figure 4.4.3: Firing rate across the τ_m gradient under increased global input strength. The normalisation rule between burst conditions is preserved, but the generator weight is increased equally for all burst conditions. Compared with the baseline regime, the intermediate burst conditions show more activity. However, the 100-burst firing-rate profile becomes less monotonic, indicating that the relationship between τ_m and firing rate is sensitive to input strength.

4.4.3 Condition-Specific Rescaling of Intermediate Burst Conditions

The third input-scaling regime tested whether activity in the intermediate burst conditions could be recovered by increasing their input strength separately. In this regime, the 10- and 100-burst conditions retained the baseline generator weight, while the 25- and 50-burst conditions were assigned the increased generator weight.

Under condition-specific rescaling, increasing the input strength for the intermediate 25- and 50-burst conditions recovered spiking activity that was weak or absent under the normalised baseline input, shown in Fig. 4.4.6, while maintaining the firing profile of for the 100-burst condition, see Fig. 4.4.5. This shows that the reduced firing in

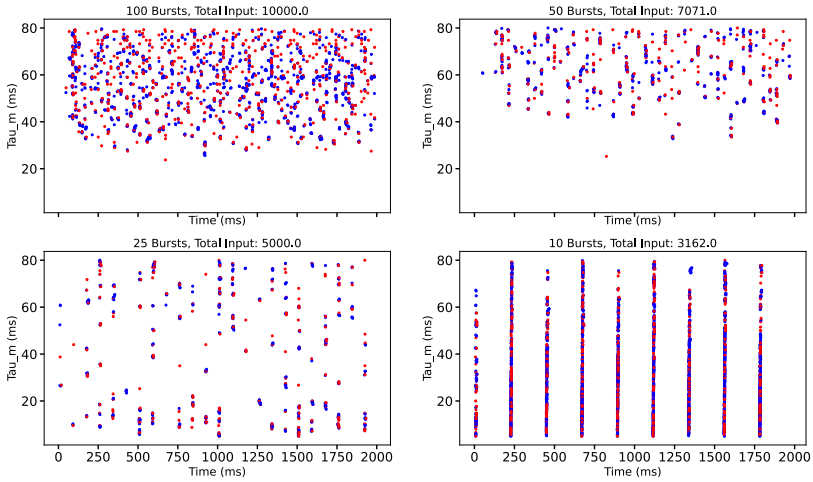


Figure 4.4.4: Spike timings under increased global input strength. The 10-burst condition produces spikes concentrated around burst times and primarily among low-τ_m neurons. The 100-burst condition produces more distributed spiking shifted toward higher τ_m values. The intermediate burst conditions show limited spiking activity.

these intermediate conditions was not caused solely by the temporal structure of the input, but also by the absolute input strength relative to the firing threshold and recurrent inhibition. However, this regime must be interpreted differently from the baseline and increased-global-input regimes. Because the input strength was no longer matched across all burst conditions, differences in firing rate can no longer be attributed only to temporal input structure. Instead, this result shows that the intermediate conditions are capable of producing activity when their input strength is increased.

The condition-specific rescaling therefore serves as an exploratory calibration rather than as the main comparable result. It demonstrates that the weak intermediate activity observed under baseline input was not caused by an inability of the network to respond to 25- or 50-burst input. Rather, it resulted from the interaction between the chosen normalisation rule, threshold, recurrent inhibition, and temporal structure of the input.

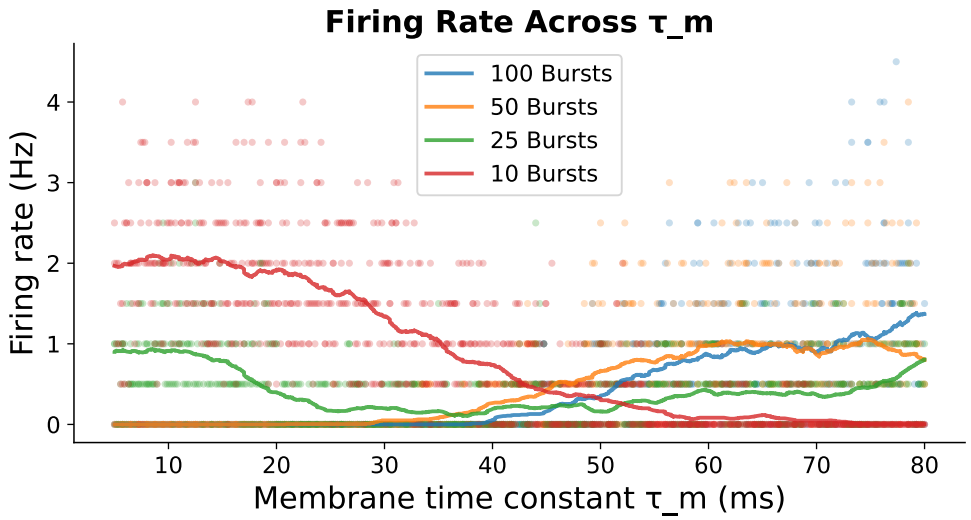


Figure 4.4.5: Firing rate across the τ_m gradient with condition-specific rescaling of the intermediate burst conditions. The 25- and 50-burst conditions receive increased input strength, while the 10- and 100-burst conditions retain the baseline input strength. This recovers activity in the intermediate conditions, but because the input strength is no longer matched across all burst conditions, this result is interpreted as an exploratory calibration rather than as the main comparable result.

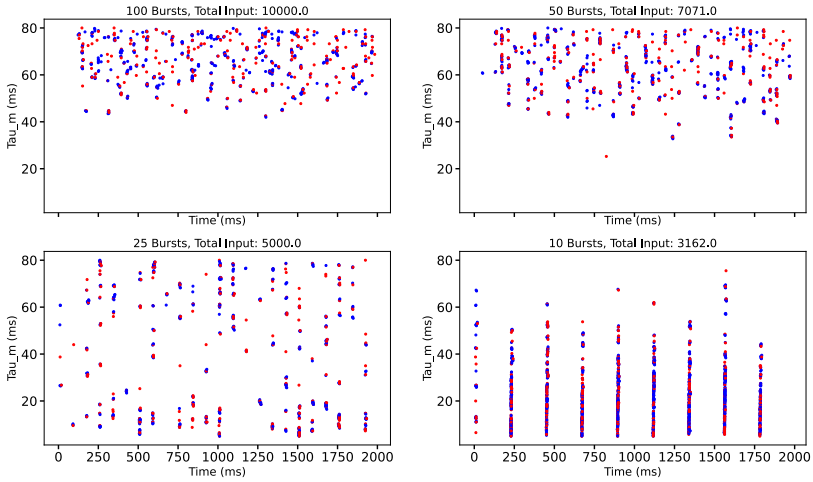


Figure 4.4.6: Spike timing across the τ_m gradient under condition-specific input rescaling. The 25- and 50-burst conditions receive increased input strength and therefore show recovered intermediate activity, while the 10- and 100-burst conditions retain the baseline input strength and preserve their low-τ_m and high-τ_m response patterns.

4.4.4 Summary Comparison

The main heterogeneous-network simulations show that temporal input structure influences which parts of the heterogeneous gradient are recruited, but that this effect is strongly dependent on input scaling. Under normalised baseline input, the clearest activity occurred in the two temporal edge conditions. The 10-burst condition preferentially activated low-τ_m neurons, while the 100-burst condition preferentially activated high-τ_m neurons. The intermediate 25- and 50-burst conditions produced weak activity under the same normalisation rule.

Increasing the global input strength recruited additional activity, including in the intermediate burst conditions. However, it also changed the firing-rate profile of the 100-burst condition, showing that stronger input altered the recurrent network dynamics rather than simply amplifying the baseline response. Condition-specific rescaling recovered activity in the intermediate conditions, but only by relaxing the normalisation constraint between burst conditions.

Together, these results suggest that the network supports τ_m/C_m-

dependent temporal selectivity, but that the observed firing-rate trends should not be interpreted as independent of input strength. Instead, temporal structure, input magnitude, recurrent inhibition, and spatial organisation jointly determine the spiking activity. The most robust conclusion is therefore not that the network produces a fixed monotonic mapping from burst number to τ_m, but that continuous heterogeneity creates input-dependent response regimes whose expression depends on the scaling of external drive.

Chapter 5

Discussion

The results suggest that continuous intrinsic heterogeneity can structure network activity by making different regions of the network responsive to different temporal input patterns. In the validation experiments, neurons with shorter membrane time constants responded more strongly to temporally concentrated input, while neurons with longer membrane time constants responded more strongly to temporally distributed input. This relationship was preserved qualitatively in the full heterogeneous network, where the 10-burst condition favoured the low-τ_m region and the 100-burst condition favoured the high-τ_m region.

This pattern suggests, but does not establish, that continuously varying neuronal properties contribute to temporal specialisation in a spatially embedded network. Rather than all neurons responding equally to the same input, the heterogeneous gradient caused different parts of the network to become more sensitive to different temporal structures. This is consistent with the idea that neural heterogeneity can influence firing-rate variability and network dynamics. However, this interpretation depends on the specific parameterisation and network configuration used, and it is not yet clear how robust this effect would be under alternative connectivity structures or different input regimes.

The results also show that this temporal selectivity was not equally strong for all burst conditions. Under the normalised baseline input,

the intermediate 25- and 50-burst conditions produced weak activity. This indicates that the relationship between burst structure and firing rate was not a simple linear progression across the gradient. Instead, the network expressed the clearest selectivity at the two temporal extremes. This raises the question of whether the observed selectivity reflects a genuine tuning of the network to temporal structure alone, and if other input conditions than temporal structure, such as input strength, affects the selectivity. The weak activity in the intermediate burst conditions can partly be interpreted using the peak-potential analysis from Section 4.2.2. Although the burst conditions were normalised according to equivalent input energy, the resulting peak membrane potentials were not equally close to threshold across the τ_m/C_m gradient. This reveals that the energy-based normalisation does not make all burst conditions equally effective at driving neurons toward the threshold. This in turn further strengthens the argument that the interpretation depend on specific configurations. The strongest peak responses occurred for the edge conditions: 10 bursts in the low-τ_m region and 100 bursts in the high-τ_m region. The 25- and 50-burst conditions produced lower peak responses across much of the gradient. This suggests that the intermediate inputs were already disadvantaged before recurrent network effects were considered. Hence, rather than strongly activating either fast-responsive or slow-integrating neurons, they fell between the two response regimes. However, this interpretation is based on derived membrane potential metrics and does not directly demonstrate suppression mechanisms within the recurrent network dynamics.

In the full connected network, recurrent inhibition and spatial connectivity may have amplified this initial difference. If an input condition brings neurons close to threshold but does not reliably trigger activity, inhibition can suppress the response before recurrent excitation is able to propagate it. This provides a possible explanation for why the intermediate conditions remained weak under the normalised baseline input. However, because this mechanism was not directly isolated in the current simulations, the observed suppression cannot be attributed to recurrent inhibition rather than reduced effective excitation.

The input-strength comparison further showed that the observed firing-rate patterns were sensitive to the absolute strength of the external drive. When the global input strength was increased, the interme-

diate burst conditions produced more activity. This indicates that the weak intermediate activity was not caused by a complete inability of the network to respond to these inputs. However, increasing global input strength also changed the 100-burst firing-rate profile, reducing the monotonic increase toward high τ_m. This suggests that stronger input changed the balance between external drive and recurrent network interactions. One possible mechanism is that neurons with shorter membrane time constants and lower capacitance reach threshold earlier under stronger input. Their early activity could recruit recurrent inhibition before high-τ_m neurons have fully integrated the distributed input. Since the highest-τ_m region is bordered by lower-τ_m regions under the periodic spatial gradient, this suppressive effect may be amplified by the network topology. This mechanism was not isolated directly, and should therefore be interpreted as a possible explanation rather than a confirmed causal result. Furthermore, the input-strength sensitivity suggests that the observed temporal selectivity is not invariant to scaling of external drive, which limits the generalisability of the findings beyond the specific input regime explored here.

The condition-specific rescaling experiment showed that intermediate activity could be recovered by increasing the input strength of the 25- and 50-burst conditions. This produced a more visually continuous progression across burst conditions, but it also broke the normalisation constraint. Therefore, this result cannot be interpreted as a controlled comparison of temporal structure alone. Instead, it shows that the intermediate burst conditions require additional input strength to become active under the chosen threshold and network parameters. This highlights a methodological trade-off between preserving controlled comparisons across conditions and achieving visually or functionally interpretable activity patterns.

A key consideration is that the heterogeneous gradient was not a pure τ_m gradient. Because membrane capacitance was parametrised as a function of τ_m, integration time and input sensitivity co-varied across the network. This made the gradient useful for producing clear differences in temporal response, but it also means that the results cannot be attributed to membrane time constant alone. The results should therefore be interpreted as effects of a combined τ_m/C_m gradient. Alternative parametrisations could produce substantially different response profiles. For example, if C_m were held constant

while τ_m varied, neurons with longer membrane time constants would integrate input over longer periods without the same compensatory change in capacitance. The resulting balance between temporal integration, leakage, and excitability could therefore differ substantially from the behaviour observed in this study. This further emphasises that the reported firing-rate patterns are specific to the chosen model formulation rather than universal properties of heterogeneous neural networks.

5.1 Limitations

These findings support the presence of input-sensitive temporal selectivity, but several modelling choices limit how broadly the results can be interpreted.

A limitation is that the network was highly simplified. The spatial topology was one-dimensional, the connectivity rules were idealised, and the model was not fitted to biological data. The simulations therefore do not directly predict activity in a specific biological neural system. In particular, the simplified topology and absence of biological constraints on connectivity likely amplify the clarity of the observed temporal selectivity, and may therefore overestimate its prominence compared to more biologically realistic settings. Instead, they show how continuous intrinsic heterogeneity can interact with spatial connectivity and recurrent inhibition in a controlled computational setting.

In addition, the model does not vary membrane time constants independently of membrane capacitance, resulting in a coupled τ_m/C_m gradient. Consequently, the observed effects cannot be attributed solely to changes in τ_m but instead reflect combined changes in integration time and excitability.

Furthermore, the results are sensitive to input normalisation and scaling. Although energy-based normalisation was applied, it did not guarantee equivalent effective drive across burst conditions, particularly for intermediate inputs. This introduces ambiguity in interpreting differences between burst regimes as purely temporal effects.

Finally, the recurrent mechanisms underlying suppression and amplifi-

cation were not explicitly isolated in the simulations. As a result, interpretations involving inhibitory mediation remain hypothetical rather than causally demonstrated.

Taken together, these limitations suggest that the observed temporal selectivity should be interpreted primarily as a proof of concept rather than evidence that similar behaviour necessarily emerges in biological neural systems. While the simulations consistently demonstrate input-sensitive activity within the chosen modelling framework, further work is required to determine whether the same principles remain under more biologically realistic assumptions.

5.2 Benefits, Ethics and Sustainability

Computational neuroscience contributes to a deeper understanding of brain function and human behaviour in both healthy and pathological conditions. The primary objective of this project is to improve understanding of neural dynamics through computational modelling, which aligns with UN Sustainable Development Goal 3, *Good Health and Well-Being*. Although the project is of a theoretical nature, its findings could, in principle, help development in artificial intelligence, neuroscience, or clinical practices.

Furthermore, reproducible research practices, including transparent reporting of models, parameters, simulation procedures, and software implementations, allow future researchers to verify, reproduce, and extend the work. Proper citation of previous research ensures that intellectual contributions are acknowledged and that scientific progress builds upon established knowledge rather than duplicating existing efforts.

5.2.1 Environmental Impact

The project relies on simulations of spatially structured and heterogeneous neural networks. Efficient implementation, careful experimental design, and avoiding unnecessary simulations reduce computational time and energy usage, thereby lowering the project's carbon footprint. The use of a simplified one-dimensional network reduced

computational requirements while still allowing the research question to be investigated. From a sustainability perspective, selecting an appropriately simple model avoids unnecessary computational resource consumption without compromising the study objectives.

Although the computational cost of the present work was modest, future studies may require substantially larger spatial networks and extensive parameter sweeps. Such studies should consider the trade-off between increased biological realism and the additional computational resources and environmental impact required. Furthermore, if future work requires larger computational resources, additional environmental considerations include the full hardware life cycle, including extraction of raw materials, manufacturing, transportation, maintenance, and responsible recycling of electronic waste. These considerations relate to UN Sustainable Development Goal 13, *Climate Action*, and Goal 12, *Responsible Consumption and Production*.

Research into SNNs may also contribute indirectly to environmental sustainability. Since SNNs have the potential to perform computations more efficiently than conventional artificial neural networks, they may reduce the energy requirements of future AI systems for both training and inference. However, rebound effects should also be considered. If the improved efficiency of SNNs leads to wider use of AI systems, or if strong SNN performance requires specialised hardware, the overall environmental impact could increase despite efficiency gains. Because the model is relatively small and exploratory, the direct environmental impact of the present work is limited. However, the sustainability relevance becomes more important if similar spatially embedded SNN simulations are scaled to larger networks or extensive parameter sweeps.

5.2.2 Social and Economic Impact

Development in artificial intelligence, neuroscience, and clinical practice raises ethical questions regarding equal accessibility to resulting technologies, as well as the possibility that insights from the project could be applied in ethically questionable ways. For instance, conclusions drawn from the project might contribute to AI systems that are insufficiently regulated or whose societal benefits are unequally distributed. Ensuring that resulting technologies are accessible and

beneficial across different populations is an important aspect of social sustainability.

Although the project does not involve personal data, methods developed here could eventually contribute to technologies with societal consequences. Responsible development therefore requires careful consideration of how such methods are applied and appropriate governance to minimise unintended misuse.

From an economic sustainability perspective, efficient simulation methods reduce computational costs and enable more effective use of available research infrastructure and public research funding.

By adhering to honesty, transparency, and responsible resource usage, this project aims to maintain high ethical and sustainability standards across environmental, social, economic, ethical, and scientific dimensions.

5.3 Use of Generative AI

Generative AI tools, were used during the thesis work as support for language refinement, restructuring of text, and formulation of explanations. The tools were primarily used to improve clarity, grammar, and readability, as well as to suggest alternative phrasings for existing sections. They were not used to independently generate scientific results, perform simulations, or make final methodological decisions. All model design, simulation work, analysis, interpretation of results, and final text selection were reviewed and carried out by the authors.

Chapter 6

Conclusions

This thesis investigated how continuously varying intrinsic neuronal properties affect spiking activity in a one-dimensional spatially embedded spiking neural network. The central research question was how continuous neural heterogeneity influences spiking activity when combined with spatial embedding and recurrent connectivity.

The results show that the heterogeneous $\tau_m/C_m gradient$ creates temporal selectivity across the network. Temporally concentrated burst input preferentially activated neurons with shorter membrane time constants, while temporally distributed input preferentially activated neurons with longer membrane time constants. This supports the idea that continuous intrinsic heterogeneity can shape how different parts of a spatial network respond to different temporal input structures.

However, the main simulations also showed that this selectivity was sensitive to input scaling. Under normalised input, the clearest activity occurred in the temporal edge conditions, while the intermediate burst conditions produced weak firing. Increasing the global input strength recruited more activity, but also changed the firing-rate profile of the distributed input condition. Condition-specific rescaling recovered activity in the intermediate conditions, but only by relaxing the normalisation constraint. Therefore, the results indicate that temporal selectivity emerges from the interaction between intrinsic heterogeneity, input structure, input magnitude, recurrent inhibition,

and spatial connectivity.

The main conclusion is that continuous neural heterogeneity can organise spiking activity according to temporal input structure, but that the resulting firing-rate patterns are not independent of model calibration. The findings should therefore be interpreted as a theoretical demonstration of input-sensitive temporal specialisation in a simplified spatially embedded network.

6.1 Future Work

The model was restricted to a one-dimensional spatial topology. This simplification allowed us to study the interaction between continuous neural heterogeneity and distance-dependent connectivity within the scope of the project. However, biological neural systems are not one-dimensional, and the results may therefore not generalise directly to two- or three-dimensional spatial networks. A natural direction for future work is to extend the model to a two-dimensional spatial embedding, where the effects of local neighbourhood structure, radial connectivity, and spatial clustering could be investigated.

Future work should also further investigate the role of input normalisation, since the present results show that energy-based normalisation reduced differences in overall input magnitude but did not make all burst conditions equally effective at driving neurons toward the threshold. In particular, alternative normalisation schemes could be compared to determine which method best isolates temporal input structure from input strength. Additional parameter experiments over inhibitory strength, connectivity range, synaptic weights, background noise, and temporal jitter would help assess the robustness of the observed temporal selectivity. Finally, validating the model against experimental neural data would help determine whether the observed input-sensitive activity patterns reflect a biologically meaningful mechanism through which spatial organisation and intrinsic neuronal heterogeneity shape neural activity.

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